FUSIONCORE v_0

Phase 1 — Exploratory Data Analysis (EDA)

Notebook 01 | Technical Deep-Dive Analysis

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1. Abstract

This investigation presents a physics-grounded exploratory analysis of the NASA C-MAPSS turbofan engine degradation dataset, conducted as the foundational validation stage of the FusionCore predictive maintenance pipeline. Four independent gate criteria are applied across 160,359 cycles of run-to-failure data spanning 709 engines to determine whether simulated sensor trajectories faithfully encode the thermodynamic and mechanical degradation behaviour of a commercial turbofan engine. All four gates pass. Temporal autocorrelation analysis formally establishes RUL estimation as a trajectory problem rather than a point-in-time classification task. Cumulative damage analysis confirms physics-consistent degradation behaviour across single-fault and dual-fault engine populations. The corpus is deemed suitable for training physics-informed prognostic models, subject to regime normalisation for multi-regime subsets and separate treatment of single-fault and dual-fault populations.

2. Taking Stock — What the Fleet Looks Like

2.1 The Corpus (Cells 7–11)

The first thing the investigation establishes is the sheer scale and structure of what we are working with. Across the four training subsets, the corpus contains 160,359 individual engine-cycle observations from 709 distinct turbofan engines, each one tracked from healthy inception through to functional failure. This is not a sampling exercise. These are complete run-to-failure trajectories — the full lifecycle of every engine, from the first cycle on the ground to the last cycle before the engine can no longer perform its mission. Crucially, the degradation encoded in each trajectory is not stochastic noise: the C-MAPSS corpus is generated by the T-MATS (Toolbox for the Modelling and Analysis of Thermodynamic Systems) simulation engine, whose degradation equations are calibrated against empirical run-to-failure data from physical turbofan rigs (Saxena & Goebel, 2008) [1]. The sensor readings are therefore thermodynamically consistent — a physically grounded proxy for the underlying degradation process.

The four subsets represent a deliberate spectrum of operational complexity:

FD001 is the controlled baseline: 100 engines, all operating at sea-level static conditions, all suffering a single fault mode — progressive High-Pressure Compressor degradation. Within the C-MAPSS simulation, HPC deterioration is implemented as a coupled reduction in compressor efficiency factor and flow capacity factor **[1]**. The thermodynamic consequences trace directly through the engine cycle: as efficiency erodes, progressively more shaft work is required to sustain the same pressure ratio, and by the standard isentropic compressor outlet temperature relationship[1]:

$$T_{30} = T_{24}\left(1 + \frac{\pi_c^{\left\{\frac{\gamma-1}{\gamma}\right\}} - 1}{\eta_{is}}\right)$$

A decline in η_{is} produces a monotonic rise in HPC outlet total temperature (s3, T30). Simultaneously, as flow capacity falls, corrected mass flow through the compressor decreases, depressing both HPC outlet total pressure (s7, P30) and static outlet pressure (s11, Ps30). The engine control system responds by increasing fuel scheduling to maintain demanded thrust, which drives a measurable rise in the fuel-flow-to-static-pressure ratio (s12, φ) and elevated bleed enthalpy (s17). Because FD001 operates at a single fixed flight condition, inlet temperature (s1, T2) and inlet pressure (s5, P2) are invariant — confirmed by the variance analysis as dead sensors — and the HPC degradation signature is uncontaminated by operating-point variation. The monotonic progressions in T30, P30, and φ constitute as clean a thermodynamic degradation signature as the dataset provides. It is the simplest possible version of the problem, and its simplicity is its value. Any model that cannot learn RUL on FD001 is not a viable predictive maintenance system.

FD002 introduces the first layer of real-world complexity. The same single HPC fault mode, but now 260 engines flying across six distinct flight regimes — from ground idle to cruise at altitude. The underlying fault physics are identical to FD001: efficiency-capacity collapse in the HPC, the same consequent thermodynamic signatures in T30, P30, Ps30, and φ. What changes is the operating-point context in which those signatures are observed. At high-altitude cruise conditions, reduced ambient temperature and pressure shift the corrected speed parameters (s13, NRf; s14, NRc) and compress the absolute dynamic range of core-side measurements. A degraded engine at altitude can produce T30 and P30 readings indistinguishable in magnitude from a healthy engine at a more thermodynamically demanding ground condition. The degradation signal must therefore be resolved against a regime-indexed baseline rather than a global reference threshold — the same physics, now masked by operating-point context. This is where the problem starts to resemble actual airline operations, and where regime-aware normalisation becomes a prerequisite rather than an optimisation.

FD003 reintroduces the controlled sea-level environment of FD001 but adds a second fault mode: simultaneous fan and HPC degradation. Fan deterioration introduces a qualitatively distinct thermodynamic signature to the signal space. As fan efficiency and flow capacity erode, bypass duct total pressure (s6, P15) falls, bypass ratio (s15, BPR) contracts, and physical fan speed (s8, Nf) departs from its healthy operating locus — parameters that are invariant in FD001 and FD002 and correctly identified as dead sensors in those single-fault subsets. The diagnostic challenge is therefore one of fault attribution: both degradation modes propagate into shared downstream indicators including core speed (s9, Nc), LPT outlet temperature (s4, T50), and bleed conditions (s17, s20, s21), and the system must correctly partition the deterioration signal between two co-degrading components. Beyond attribution, the two fault modes interact physically: fan degradation reduces the total mass flow presented to the LPC and HPC inlet stages, which can accelerate the compressor stall margin deterioration already imposed by HPC efficiency loss. This coupling means the compound fault is not

[1] Isentropic refers to a thermodynamic process that is both adiabatic (no heat transfer to or from surroundings) and reversible — it represents the theoretical ideal limit of compressor performance. The relationship defines the outlet total temperature a perfect compressor would produce for a given pressure ratio. A real compressor is less efficient than this ideal, so the actual outlet temperature is higher. This is the direct physical mechanism behind the monotonic EGT rise observed in s4 and the s3 (T30) signal in the C-MAPSS data.

simply additive in severity — deterioration rates can exceed those of either fault in isolation. The sea-level operating condition at least ensures that the sensor signals carry no additional regime-dependent confounding, isolating the attribution problem in its purest form.

FD004 is the most demanding subset: 249 engines, six flight regimes, two fault modes simultaneously. The full diagnostic complexity of FD003 — compound fan and HPC degradation, cross-coupled thermodynamic signatures, attribution ambiguity between co-degrading components — is superimposed on the operating-point variability of FD002. At altitude, reduced ambient conditions amplify the relative sensitivity of bypass-side parameters to fan degradation while simultaneously compressing the absolute dynamic range of core-side HPC signals, making both fault modes harder to resolve independently and harder to attribute jointly. This represents the real-world operational environment in its full complexity, and the ultimate validation target for any predictive maintenance system aspiring to operational deployment.

Table 1 — Fault Mode Thermodynamic Signatures and Sensor Diagnostic Role by Subset

Sensor	Physical Quantity	HPC Only (FD001 / FD002)	HPC + Fan (FD003 / FD004)	Regime Sensitivity
s3	T30 — HPC outlet temperature	↑ monotonic; primary degradation indicator	↑ compounded by core mass flow reduction	Low (single); **High (multi)**
s7	P30 — HPC outlet total pressure	↓ monotonic; flow capacity collapse	↓ exacerbated by reduced inlet delivery	Low; **High**
s11	Ps30 — HPC outlet static pressure	↓ co-monotonic with P30	↓ amplified	Low; **High**
s12	φ — fuel-flow / Ps30 ratio	↑ FADEC compensation signal	↑ further elevated by compound thrust loss	Low; **High**
s4	T50 — LPT outlet temperature (EGT)	↑ elevated through turbine stages	↑ compounded across both fault modes	Low; **High**
s17	htBleed — bleed enthalpy	Active; shifts with HPC operating point	Active; altered by both components	Low; **High**
s6	P15 — bypass duct total pressure	**Invariant** — dead sensor in FD001/FD002	↓ fan flow capacity loss — becomes active	Low; altitude amplifies sensitivity
s15	BPR — bypass ratio	**Invariant** — dead sensor in FD001/FD002	↓ bypass flow contraction	Low; **High**
s8	Nf — physical fan speed	Minor operating point shift	Departure from healthy locus	Low; **High**
s9	Nc — physical core speed	Shift with compressor map migration	Shared indicator for both fault modes	Low; **High**

2.1.1 Key observations for the record:

The dead-sensor confirmation from Phase 1 variance analysis is not a data quality artefact — it is physically correct. In FD001, the bypass-side sensors (s6, s15) carry no degradation information because HPC-only deterioration does not appreciably disturb fan operating conditions at a fixed sea-level point. The moment fan degradation is introduced in FD003, s6 and s15 re-emerge as active diagnostic channels. This is the variance analysis doing exactly what it should: revealing which thermodynamic pathways are being exercised by the fault mode in question.

The regime effect is equally physically grounded. Corrected parameters (s13, s14) are more thermodynamically stable across flight conditions than their physical counterparts (s8, s9) precisely because they normalise for inlet temperature and pressure — a fact that motivates the K-Means regime clustering approach mandated for Phase 2 normalisation.

2.2 The Lifespan Story — And What It Tells Us About Fault Severity

The engine lifespan summary (Table 2) produced in Cell 9 contains more diagnostic information than its compact form suggests.

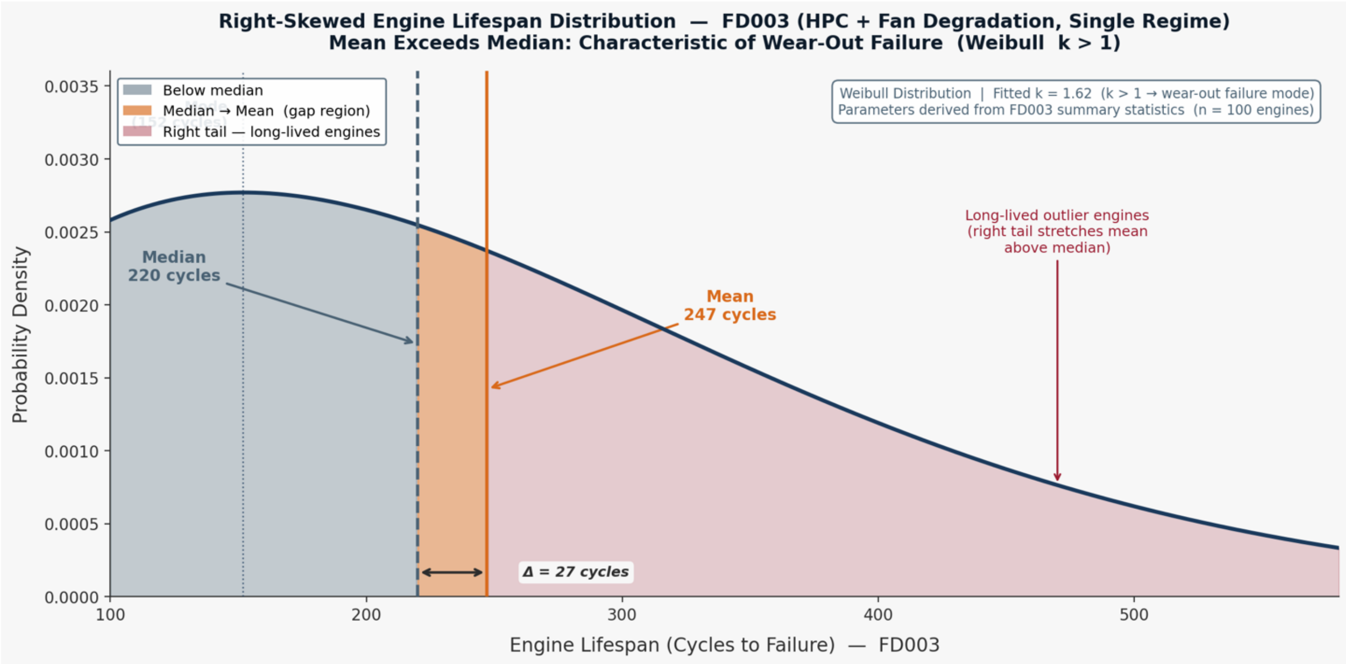
Table 2: Engine Lifespan Summary

Subset	Engines	Min Cycles	Median	Max	Mean
FD001	100	128	199	362	206.3
FD002	260	128	199	378	206.8
FD003	100	145	220	525	247.2
FD004	249	128	234	543	246.0

The most immediately striking observation is the persistent ~40-cycle mean lifespan difference between the single-fault and dual-fault subsets. FD001 and FD002 converge on a mean of approximately 206–207 cycles, irrespective of operating complexity. FD003 and FD004 converge on approximately 246–247 cycles. The difference is not noise — it is mechanically meaningful. Fan degradation, as simulated by the C-MAPSS model, progresses more slowly than HPC degradation alone. An engine suffering fan deterioration survives, on average, 40 additional cycles. In the context of real airline operations, where a single engine cycle might represent a short-haul flight segment, this is the equivalent of four to six additional weeks of revenue-generating service before intervention becomes necessary.

The maximum lifespan values are equally instructive. FD003 produces engines surviving up to 525 cycles and FD004 up to 543 cycles — more than double the median. Examining the distribution shape reveals that the mean lifespan consistently exceeds the median across both subsets: in FD003, the mean of 247.2 cycles sits above the median of 220 cycles; in FD004, the mean of 246.0 cycles sits above the median of 234 cycles. This divergence arises because a subset of long-lived engines pulls the arithmetic mean upward whilst the median — the true midpoint of the fleet — remains anchored closer to the typical engine. The result is a right-skewed distribution, illustrated in Figure 1, in which the bulk of the fleet clusters around a modal lifetime but a meaningful right tail extends to engines living considerably longer.

Figure 1. The 27-cycle gap between mean and median confirms that a meaningful minority of engines outlive the typical fleet — making fixed-interval maintenance policies both wasteful and potentially unsafe.



This distributional shape is characteristic of a Weibull[2] [2] failure time model with a shape parameter $k > 1$. Under the Weibull framework – *where the distribution is the standard statistical model for engineering component lifetimes*; the shape parameter governs how the probability of failure changes with age. At $k = 1$, failures occur at random, independent of how long the component has operated — as one might observe in electronics failures from random voltage surges. At $k > 1$, the probability of failure in any given cycle *increases the longer the engine has already operated*. This is the mathematical definition of a wear-out failure mode, and it is the physically expected behaviour of thermomechanical components — compressor blades, turbine discs, and seals — that accumulate fatigue damage irreversibly with every cycle of thermal and mechanical loading. In plain terms: the longer an engine runs, the faster its probability of failure grows. This is precisely the behaviour that makes predictive maintenance commercially valuable. A purely time-based maintenance programme (replace at fixed intervals) either replaces engines far too early, wasting serviceable life, or — more dangerously — misses the long-tailed engines that deteriorate rapidly in their final cycles.

2.3 Business Impact

The lifespan variability observable in Table 2 is the commercial case for predictive maintenance in a single number. An airline operating a fleet of 249 engines with lifespans ranging from 128 to 543 cycles cannot optimise maintenance intervals using any fixed-interval policy. A model that can distinguish a 200-cycle engine from a 400-cycle engine with comparable current sensor readings is worth, conservatively, tens of millions of pounds in avoided early removals and AOG events annually.

[2] A mathematical based explanation is given in A.1.

3. The Physics Audit (Cells 13–31)

3.1 Descriptive Statistics — Reading the Fleet's Vital Signs

Before examining which sensors carry useful information, the descriptive statistics table produced in Cell 13 establishes whether the data's raw values are physically plausible. This is the foundational credibility check: a machine learning model trained on physically implausible sensor readings would learn nothing transferable to real engines.

The FD001 statistics tell a compelling story of physical coherence. The T2 sensor (s1, fan inlet temperature) reports a perfectly stable mean of 518.67 °R[3] — 288.15 K, which is precisely ISA standard sea-level ambient temperature. This is not coincidence; it is the simulation confirming that a stationary engine at sea level breathes standard atmosphere. The T50/EGT sensor s4 shows a mean of approximately 1,409 °R (~783 K, ~510 °C) with a standard deviation of around 9 °R — a physically plausible exhaust gas temperature at take-off power. The HPC outlet pressure (s7, P30) sits at a mean of 553 psia, consistent with a high-pressure ratio turbofan at full thrust.

The transition to FD002 descriptive statistics reveals the multi-regime effect in full. s1 (T2) now shows a mean of 472.9 °R with a standard deviation of 26.4 °R — whereas in FD001 it was constant to six decimal places. s7 (P30) expands from a tight sea-level distribution to a range spanning 136.8 to 555.8 psia, with a standard deviation of 146 psia. This is not measurement noise. This is altitude. At cruise altitude, ambient pressure is lower, so HPC outlet pressure is lower; at sea-level take-off, it is higher. The sensor is physically alive and reporting correctly — it is simply now reporting a mixture of flight conditions rather than a single one.

This distinction — the same sensor being informationally useless in FD001 and informationally rich in FD002 — is one of the most important results in the entire EDA, and it has profound implications for any cross-subset model.

3.2 The Variance Audit — Finding the Dead Sensors (Cells 23–26)

The variance audit, executed across all four subsets with a detection threshold of $\tau = 10^{-5}$, produces the single most consequential table in the investigation. For FD001, the ranked variance output is:

3.2.1 Dead Sensors (FD001/FD003):

1. s18 (Demanded Fan Speed) = 0.000,
2. s19 (Demanded Corrected Fan Speed) = 0.000,
3. s16 (Fuel-Air Ratio) = $2.42 \times 10^{(-28)}$,
4. s10 (Engine Pressure Ratio) = 2.17×10^{-25},
5. s5 (Fan Inlet Pressure) = 1.15×10^{-23},
6. s1 (Fan Inlet Temperature) = 4.27×10^{-21},
7. s6 (Bypass Duct Pressure) = 1.93×10^{-6}

[3] Degrees Rankine (°R) — the absolute thermodynamic temperature scale referenced to absolute zero on the Fahrenheit interval, defined as T(°R) = T(°F) + 459.67. The conversion to SI units is T(K) = T(°R) / 1.8. The C-MAPSS dataset adopts Rankine throughout all temperature channels (s1 T2, s3 T30, s4 T50) in accordance with the US aerospace engineering convention of the underlying T-MATS simulation [Saxena & Goebel, 2008]. The nominal sea-level ISA standard day ambient temperature of 518.67 °R corresponds to 288.15 K (15 °C), confirming the fixed inlet condition modelled in FD001 and FD003.

Table 3. Establishing the dead sensors from the subsets.

Sensor	FD001	FD002	FD003	FD004
s1	4.273435e-21	6.964167e+02	4.468817e-21	6.989061e+02
s2	2.500533e-01	1.390499e+03	2.735616e-01	1.394473e+03
s3	3.759099e+01	1.122463e+04	4.638179e+01	1.127156e+04
s4	8.101089e+01	1.419039e+04	9.551501e+01	1.423907e+04
s5	1.152399e-23	1.305983e+01	1.297818e-23	1.312520e+01
s6	1.929279e-06	2.950447e+01	3.281895e-04	2.963732e+01
s7	7.833883e-01	2.131755e+04	1.181533e+01	2.157380e+04
s8	5.038938e-03	2.108589e+04	2.505412e-02	2.112611e+04
s9	4.876536e+02	1.127697e+05	3.992122e+02	1.135202e+05
s10	2.172333e-25	1.624831e-02	1.214417e-05	1.630232e-02
s11	7.133568e-02	1.044823e+01	9.004450e-02	1.052024e+01
s12	5.439850e-01	1.895014e+04	1.059707e+01	1.917646e+04
s13	5.172330e-03	1.640148e+04	2.500215e-02	1.643469e+04
s14	3.639005e+02	7.197478e+03	2.723859e+02	7.339442e+03
s15	1.406628e-03	5.615037e-01	3.661655e-03	5.630607e-01
s16	2.422479e-28	2.219799e-05	3.063800e-28	2.194647e-05
s17	2.398667e+00	7.703131e+02	3.102736e+00	7.733006e+02
s18	0.000000e+00	2.112022e+04	0.000000e+00	2.116225e+04
s19	0.000000e+00	2.877321e+01	0.000000e+00	2.883071e+01
s20	3.266927e-02	9.740370e+01	6.193364e-02	9.873197e+01
s21	1.171825e-02	3.506553e+01	2.227072e-02	3.555376e+01

These seven sensors — s1, s5, s6, s10, s16, s18, s19 — carry essentially zero variance in FD001 and FD003. They are flat lines. They are not broken; they are simply reflecting the physics of a single operating condition, namely ambient temperature and ambient pressure at Sea Level[4]. Fuel-air ratio is held constant by the engine control system at a fixed thrust setting. Engine pressure ratio is a function of the operating point, which is fixed.

[4] This can be observed in A.2 Sensor Variance Across Subsets

The critical finding is that this seven-sensor dead pattern matches the published literature exactly, as confirmed by the Gate 2 verification. This is the investigation's first formal claim: the C-MAPSS data is a faithful representation of known turbofan physics.

The active sensors in FD001 are equally instructive in their ranked variance:

3.2.2 Highest Variance Active Sensors (FD001):

- s9 (Core Speed, Nc) = 487.7,
- s14(Corrected Core Speed, NRc) = 363.9,
- s4 (EGT/T50) = 81.0,
- s3 (HPC Outlet Temperature, T30) = 37.6

The sensors with the highest variance are those whose values change as the engine degrades — not because of noise, but because degradation physically alters the thermodynamic state of the gas path. The core speed and corrected speeds rise as HPC efficiency falls (the control system commands higher shaft speed to compensate for reduced compression work). The EGT rises as the combustor must work harder for the same thrust. This ranked variance table is not just a statistical curiosity — it is a physics-validated ranking of diagnostic importance – which has been graphically illustrated in Figure 2.

The contrast with FD002 is stark and instructive. In FD002, there are zero dead sensors. Every sensor that was frozen in FD001 springs to life the moment the flight envelope is opened. s18 (Demanded Fan Speed) jumps from variance of 0 to 2.11×10^4 — a transition from effectively zero to substantial variance representing a regime-induced variance inflation factor exceeding 10^{29}**.** This is the mathematical statement of the multi-regime confounding problem. A model that does not separate regime-driven variance from degradation-driven variance will learn a corrupt representation of engine health.

3.3 Business Impact

The identification of seven dead sensors in a single-regime operation has a direct consequence for sensor instrumentation strategies. An airline operating in a narrow flight profile (short-haul, single runway, consistent thrust settings) may find that certain on-wing sensors add cost without diagnostic return. More immediately, any real-time health monitoring system deployed on narrowly-operated fleets must normalise for operating condition before drawing health conclusions from sensor trends — or it will trigger false alerts on healthy engines merely transitioning between altitude bands and miss genuinely degrading engines whose regime variation masks the deterioration signal.

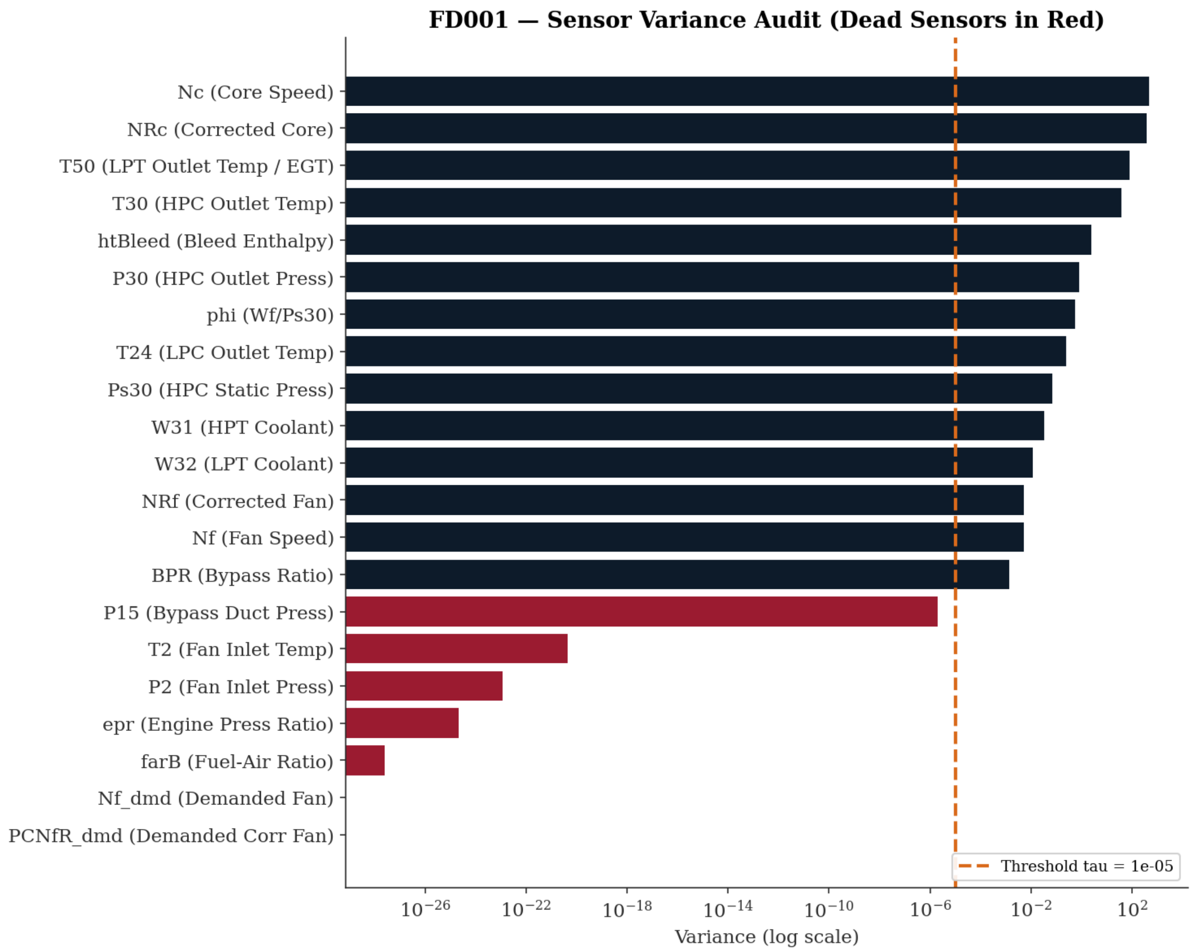
3.4 KDE Validation — The Shape of Health and Disease (Cells 28–30)

The Kernel Density Estimation plots produced in Figure 3 provide visual confirmation of what the variance table expressed numerically. The 2×2 comparison panel contrasts dead versus active sensors in both single-regime and multi-regime contexts.

In the top-left panel, s1 (T2) in FD001 renders as a near-perfect Dirac spike — an infinitely narrow distribution concentrated at 518.67 °R. The distribution has essentially no width. This is the visual signature of a constant. From a model's perspective, this sensor offers no discriminatory information whatsoever; every observation at every health state reports the same value.

The top-right panel shows s4 (EGT/T50) in FD001 as a Gaussian distribution with mild right skew, centred near 1,409 °R with a spread of approximately ± 20 °R visible in the distribution body. This is the visual signature of a degradation-sensitive, noise-perturbed continuous measurement. The slight right skew is physically meaningful: as engines degrade, EGT is pulled towards higher values, stretching the right tail of the distribution.

Figure 2. Horizontal bar chart depicting the 7 dead sensors and the highest variance active sensors from subset FD001.



The bottom-left panel is perhaps the most analytically interesting. It shows s1 (T2) in FD002 — the same sensor that appeared as a Dirac spike in FD001, now transformed into a multimodal distribution with at least four distinct peaks. Each peak corresponds to one of the six flight regimes, at their respective altitude-dependent ambient temperatures: ground level (~518 °R), low altitude (~491 °R), medium altitude (~462 °R), high altitude (~449 °R), and so on. This multimodal KDE is the visual proof that T2 in FD002 is not reporting engine degradation — it is reporting altitude. Any regression model receiving raw T2 values from FD002 is receiving altitude data mislabelled as engine health data.

The bottom-right panel shows s4 (EGT/T50) in FD002 — also multimodal, but with each mode broader than the corresponding FD001 distribution. The breadth within each mode reflects both noise and degradation; the separation between modes reflects altitude. A further demonstration of an active -vs-dead sensor, given a dataset/regime is illustrated in Figure 4.

Figure 3. Variances of 2 selected sensors under different regimes.

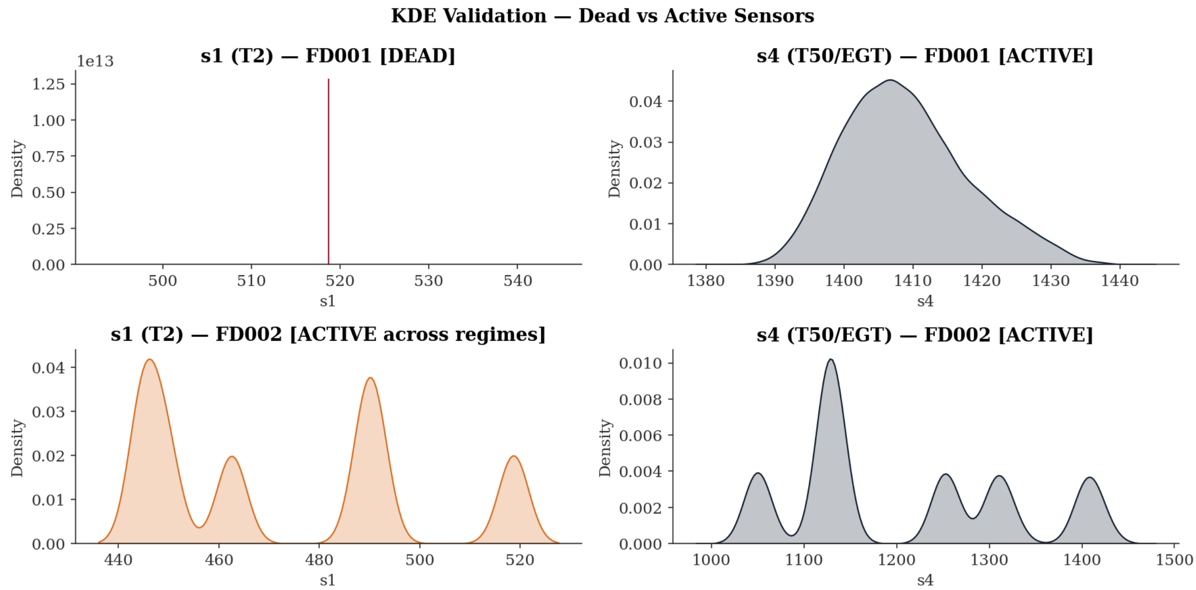
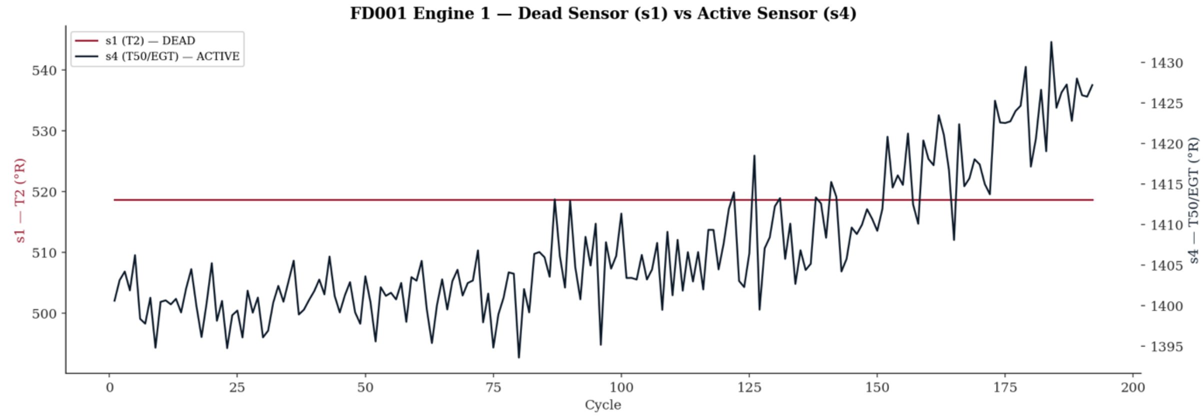


Figure 4. A direct comparison between s1 (dead) and s4 (active) sensors



4. What Time Tells Us — Temporal Memory and Serial Correlation (Cells 32–41)

4.1 ACF and PACF — The Engine's Memory

The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots, computed for what are referred to here as the primary fatigue sensors (s4, s7, s9) — a deliberate misnomer used for brevity, where "fatigue" denotes not structural fatigue in the mechanical sense but rather the sensors that exhibit some of the strongest, most consistent response to cumulative engine degradation across the lifecycle — represent the formal statistical characterisation of time dependence in the degradation signal. In plain terms, when EGT, HPC outlet pressure, and core speed are measured cycle after cycle across an engine's life, each reading is not independent of the one before it. If EGT is slightly elevated today, it will very likely still be slightly elevated tomorrow — because whatever caused it to rise has not gone away. Degradation is a slow, continuous, accumulating process, not a random event. The ACF quantifies how similar a sensor reading is to its own reading one cycle ago, five cycles ago, thirty cycles ago — and for a degrading engine, the answer is: very similar, for a long time. That slow decay in similarity is the statistical fingerprint of a process that carries memory. The PACF answers a more refined question: once the reading at lag 1 is accounted for, does lag 2 add any further explanatory power? For these sensors the answer is essentially no — lag 1 explains almost everything, meaning the single most important piece of information about where the engine is today is where it was yesterday. Taken together, these two plots formally prove that sensor readings are a connected chain of events through time, not a series of isolated snapshots — and that proof is what justifies building a model that uses trajectory history rather than the current reading alone.

The ACF for s4 (EGT) reveals slowly decaying autocorrelation that remains above the statistical confidence band to approximately lag 17, beyond which it drops within the confidence region. At lag 1 the autocorrelation is approximately 0.8–0.85, decaying gradually but remaining statistically significant across the first 17 lags. s7 (P30) exhibits a broadly similar pattern. s9 (Nc), by contrast, decays sharply — dropping to within the confidence bands by approximately lag 9–10 and remaining there. This persistent but finite memory in s4 and s7 is the statistical fingerprint of a non-stationary[5] integrated process — the sensor reading at any given cycle is highly predictable from recent history, and recent history is itself predictable from the cycle before. In physical terms, EGT and HPC outlet pressure do not jump randomly between cycles; degradation accumulates gradually and continuously, and the thermodynamic state of the engine today is strongly conditioned on its state yesterday. The PACF for s4 and s7 tells the complementary story: significant partial autocorrelation extends to approximately lag 3–4 before dropping within the confidence bands, confirming that whilst lag 1 carries the dominant temporal dependency, the immediately preceding two to three cycles also contribute meaningful predictive information, Figure 5.

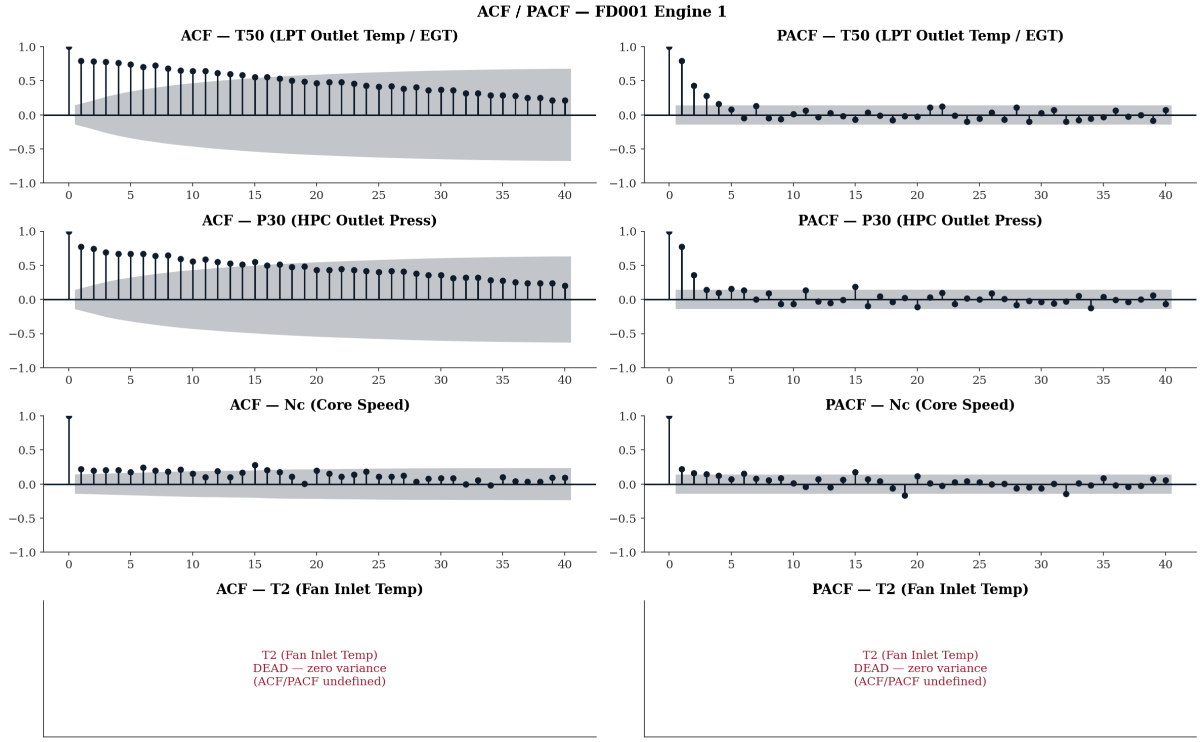
This lag-1 PACF dominance is the formal statistical justification for including the first-order difference of sensor readings (the cycle-to-cycle rate of change) as a feature in the FusionCore modelling pipeline. The first difference essentially captures how rapidly the engine's condition is evolving — the velocity of degradation rather than its absolute level. A high positive first difference in EGT (EGT rising rapidly per cycle) is a stronger early warning signal than the absolute EGT value alone, because it indicates accelerating deterioration rather than a merely elevated steady state.

4.2 Business Impact

The ACF analysis establishes that this is a memory problem — the engine's current health state depends on its history. This fundamentally differentiates predictive maintenance from conventional anomaly detection. A system that assesses each sensor reading in isolation (as many simple threshold-based alerting systems do) will systematically underestimate risk for rapidly-degrading engines, because those engines' individual point readings may not yet breach alert thresholds even as their trajectory is clearly accelerating toward failure. A history-aware model, justified by the ACF evidence, can detect this acceleration early.

[5] **Non-stationarity** — a time series is strictly stationary if its statistical properties (mean, variance, autocovariance) are invariant under temporal translation. A degrading turbofan sensor channel violates this condition by definition: the process mean drifts monotonically as the engine deteriorates, and the variance structure evolves with the degradation state. The slowly decaying ACF observed in s4 and s7 — remaining significant to lag 17 — is the classical diagnostic signature of an integrated process (order I(1) or higher), in which shocks to the series are permanent rather than transient. This contrasts with a covariance-stationary process, whose ACF decays to zero geometrically. The practical implication for modelling is that differencing or regime-indexed normalisation is required before applying any method that assumes a stable mean — failure to account for non-stationarity will cause a model to conflate operating-point variation and long-run drift with true degradation signal, inflating apparent predictive skill whilst undermining generalisation across engine populations.

Figure 5. ACF and PACF for sensors s4, s7, s9 and s1.



4.3 Seasonal Decomposition — Separating Trend from Noise

The additive decomposition of s4, s7, and s9 across the lifecycle of an FD001 engine separates each sensor's time series into four components: the observed signal, its long-term trend, any periodic component, and the residual noise.

The trend components across the three sensors tell three distinct stories.

- For s4 (EGT), the trend is flat from cycles 0 to approximately 75 — the engine operates in a thermodynamically stable early-life phase with negligible drift, Figure 6. From cycle 75 onwards the trend rises progressively, accelerating through the final third of the lifecycle to reach approximately 1,425 °R by cycle 185 against an initial value of approximately 1,400 °R — a total rise of around 25 °R. This gradual-then-accelerating profile is the thermodynamic signature of progressive HPC efficiency loss: the control system initially compensates without visible thermal consequence, but beyond a degradation threshold the compensation capacity is exhausted and EGT climbs openly.
- For s7 (P30), the trend moves in the opposite direction — declining from approximately 554 psia at inception to around 552 psia by cycle 185, with the rate of decline steepening noticeably from approximately cycle 150 onwards, Figure 7. This is physically consistent with progressive HPC efficiency loss reducing the compression ratio and therefore outlet pressure. The seasonal component for s7 is extremely small in amplitude (± 0.2 psia), and the residual (± 1 psia) remains modest relative to the trend range — making s7 the cleanest degradation signal of the three.
- For s9 (Nc), the trend is not monotonically decreasing as one might expect. It rises very slightly from approximately 9,050 rpm to a local peak near cycle 50 before declining to approximately 9,044 rpm by cycle 185, Figure 7. The total trend movement across the lifecycle is therefore around 8 rpm. Notably, the residual for s9 oscillates at approximately ± 10 rpm — larger than the trend signal itself. This means that for Nc, the noise-to-signal ratio is unfavourable: the degradation trend is present but weak relative to cycle-to-cycle variability, making s9 a less reliable standalone prognostic channel than s4 or s7.

Figure 6. The deconstruction of the sensor s4 signal into its three components, Trend, Seasonality and Residual.

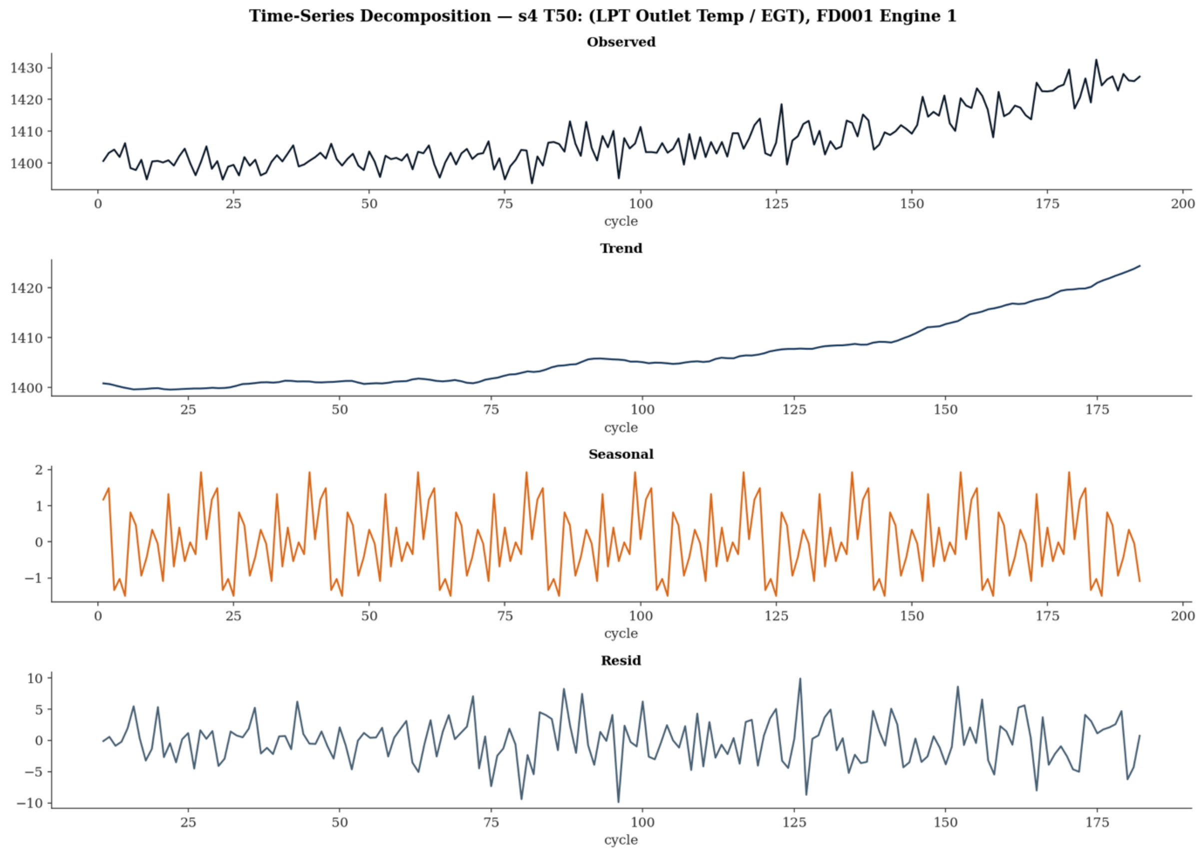
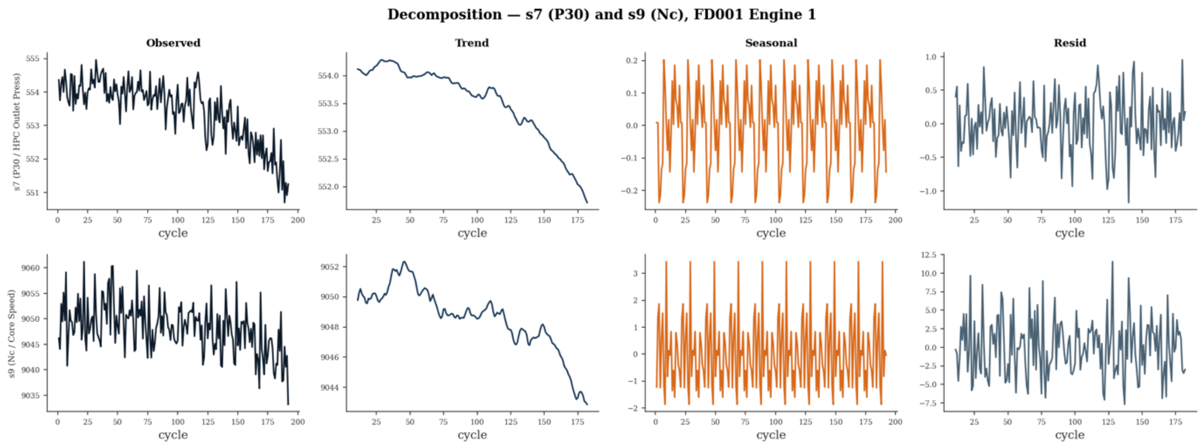


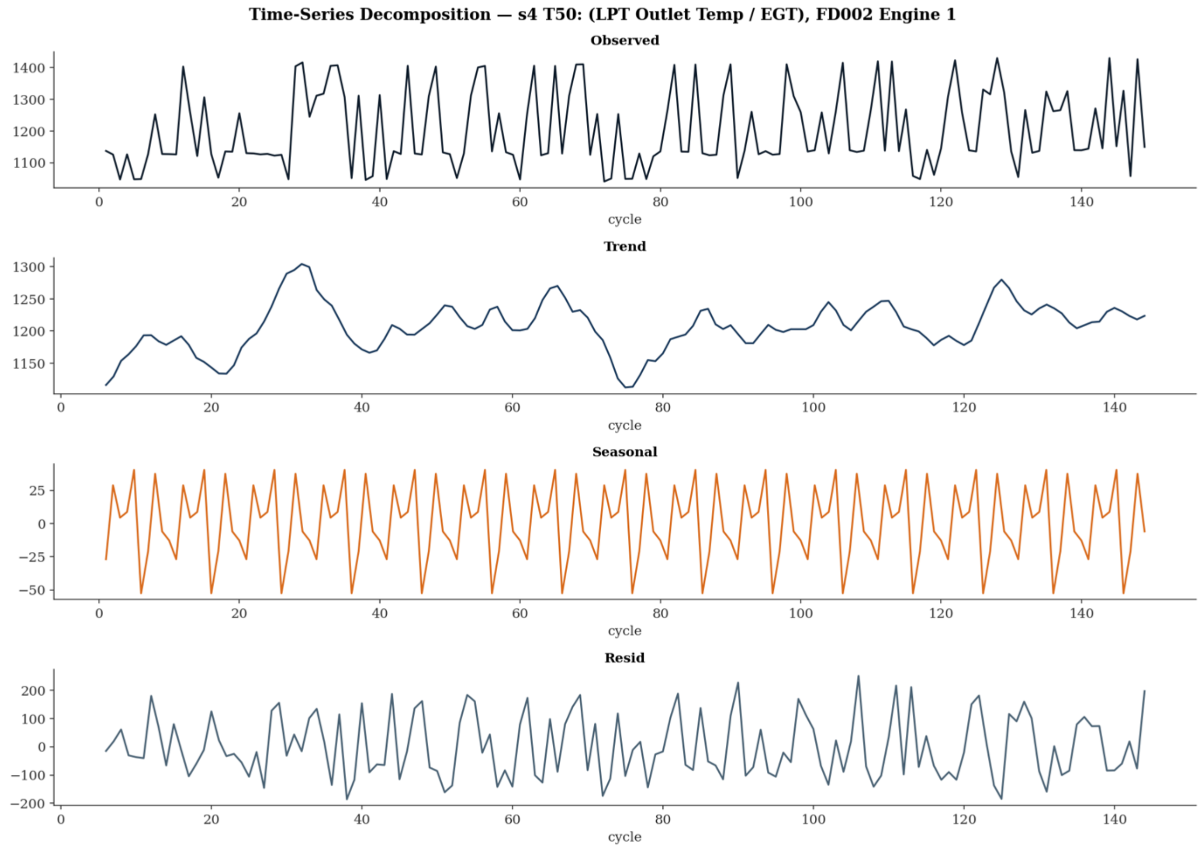
Figure 7. The deconstruction of the sensors s7 and s9 signals into their three components, Trend, Seasonality and Residual.



This directional divergence is not coincidental — it is a physically coherent consequence of HPC efficiency loss. As the compressor degrades, it delivers less compression work per unit of shaft power. To maintain the required thrust, the fuel control system increases fuel flow. The additional combustion raises the gas temperature entering the turbine, which is what the rising s4 trend records — a 25 °R increase from approximately 1,400 °R to 1,425 °R across the lifecycle. The increased fuel flow does not, however, recover the lost compression ratio; s7 (P30) declines from approximately 554 psia to 552 psia because the HPC is physically delivering less pressure regardless of how hard the control system drives it. s9 (Nc) declines by approximately 6 rpm across the same period, reflecting the control system's net inability to sustain shaft speed as mechanical efficiency falls. The three sensors are therefore not independent channels — they are three observable consequences of the same underlying degradation event, and their simultaneous divergence in these specific directions is the quantitative fingerprint of HPC deterioration in the gas path.

For FD002 (Figure 8), the decomposition exposes a fundamental problem. The observed signal oscillates between approximately 1,100 °R and 1,400 °R — a 300 °R range driven entirely by regime switching between altitude bands. The decomposition fails to isolate a clean degradation trend: the trend component itself oscillates between approximately 1,140 °R and 1,300 °R with no coherent direction, meaning regime variation has contaminated what should be a monotonic degradation signal. The residual reaches ±200 °R — twenty times larger than the equivalent FD001 residual — confirming that the overwhelming majority of variance in this signal is regime-driven rather than degradation-driven. This is the clearest possible demonstration that a decomposition applied to raw FD002 data extracts regime behaviour, not engine health. Regime normalisation must precede any feature engineering — not as a technicality, but because without it the degradation signal is unrecoverable from the noise floor.

Figure 8. The deconstruction of the sensor s4 signal into its three components, Trend, Seasonality and Residual.

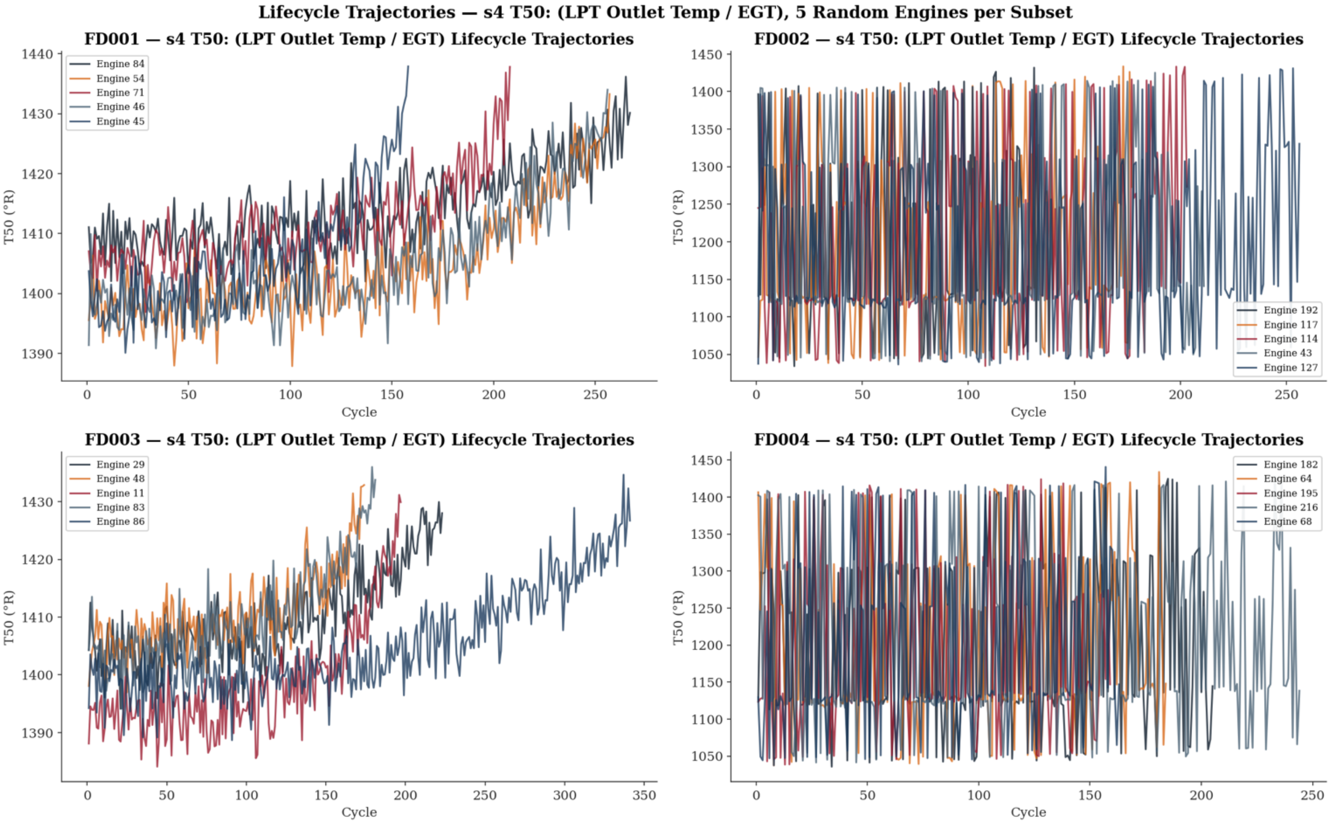


5. How a Turbofan Actually Dies — Lifecycle Trajectories and Cumulative Damage (Cells 42–51)

5.1 Lifecycle Trajectories — Five Engines, One Story

The lifecycle trajectory plots for s4 (T50/EGT), Figure 9 and s7 (P30), Figure 10, showing five randomly selected engines per subset, are the most directly interpretable output in the entire investigation from both a physical and commercial perspective.

Figure 9. Lifecycle trajectories recorded on s4 sensor on all the subsets across 5 randomly chosen engines.



FD001. The five EGT trajectories share a consistent structural pattern: a broadly flat early-life phase where readings oscillate around 1,395–1,405 °R with no directional drift, followed by a gradual upward trend that begins at varying points across the lifecycle, culminating in terminal values of approximately 1,425–1,440 °R. The total EGT rise from inception to failure is approximately 20–35 °R across the five engines. Engine lifespans vary visibly — the shortest trajectory ends at approximately cycle 155 whilst the longest extends to approximately cycle 270 — yet all five trace the same degradation direction in the same thermodynamic language. This population-level consistency is the foundation of prognostic modelling: if all engines degrade in the same direction, a model trained on 100 engines can learn the trajectory of the 101st. In the context of real commercial operations, this 20–35 °R EGT rise represents the erosion of EGT margin — the gap between current EGT and the engine's certified red-line limit. Airline EGT margin management programmes actively track this erosion, and an operator who can forecast when margin will reach a defined intervention threshold can schedule a shop visit proactively rather than responding to an in-service exceedance.

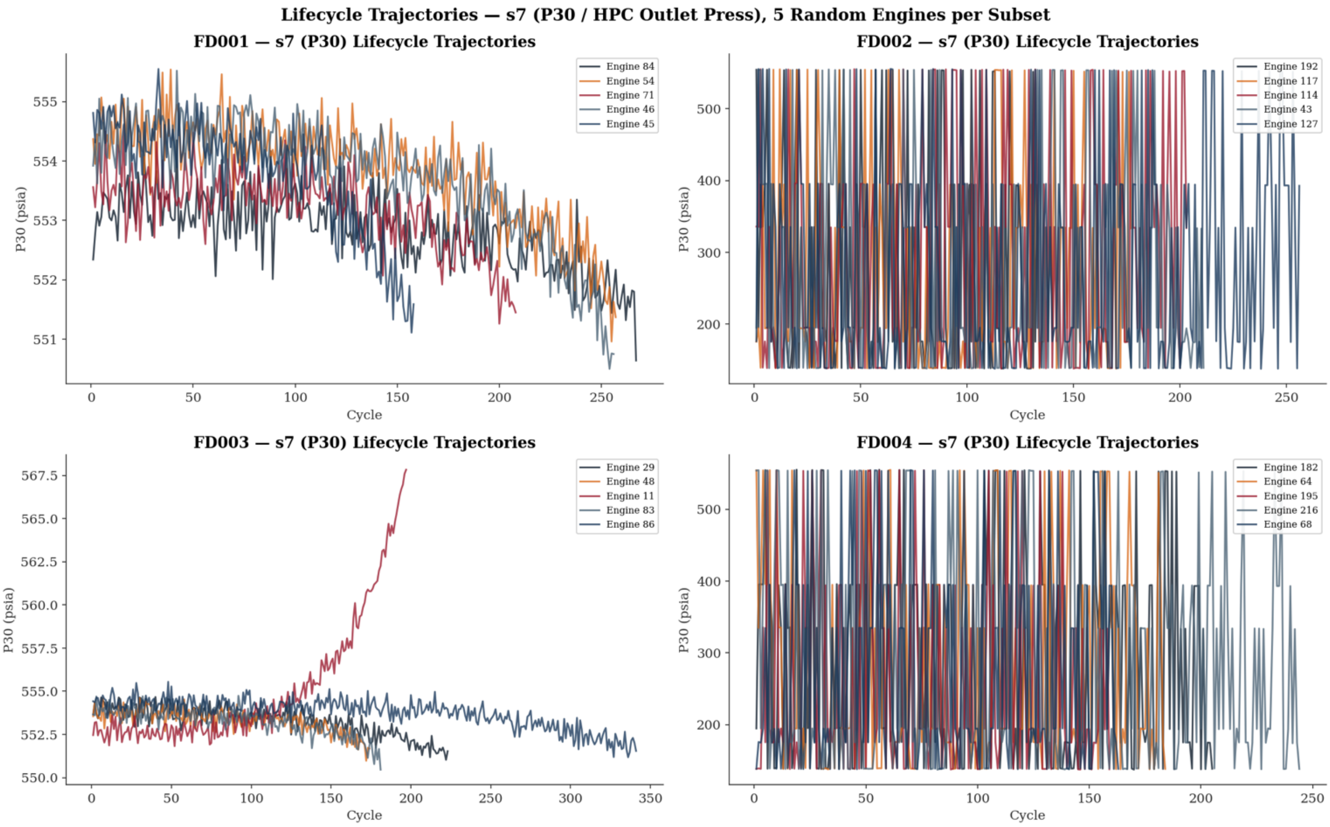
The FD001 s7 (P30) trajectories (Figure 10) tell a physically coherent but directionally opposite story. All five engines show a consistent monotonic decline from approximately 554–555 psia at inception to approximately 551–552 psia at end of life. There are no initial rise and no plateau — the decline is present from the earliest cycles and continues through to failure. This is physically consistent with the EGT rise: as HPC efficiency degrades, compression ratio falls, and P30 declines accordingly. The two sensors are not independent channels reporting conflicting information — they are complementary observables of the same underlying deterioration.

FD002 and FD004. The multi-regime subsets present an immediately different picture. Both EGT and P30 traces oscillate across the full flight envelope range — EGT spanning approximately 1,050–1,450 °R and P30 spanning approximately 150–550 psia — driven entirely by regime transitions between altitude bands. No degradation trend is recoverable from visual inspection of the raw traces. This is not a modelling challenge; it is a data preparation prerequisite. Any feature extracted from raw FD002 or FD004 sensor values without prior regime normalisation will reflect altitude rather than engine health.

FD003. The single-regime dual-fault trajectories broadly replicate the FD001 EGT pattern — flat early life followed by terminal rise — but with two notable differences. First, engine lifespans extend further, with several trajectories running to approximately cycle 300–350, consistent with the slower dual-fault degradation progression identified in the lifespan analysis. Second, inter-engine variability in degradation rate is visibly greater than in FD001: some engines maintain the flat early-life phase to well beyond cycle 150 before rising sharply, whilst others begin rising earlier and more gradually. The P30 trajectories in FD003 reveal an additional observation absent in FD001: Engine 11 exhibits an anomalous mid-lifecycle rise from approximately 552 psia to 567 psia before terminating — behaviour inconsistent with the monotonic decline seen across all other engines in both FD001 and FD003. Whether this reflects a simulation artefact or a physically distinct fault progression pathway warrants scrutiny before this engine's trajectory is used in model training without qualification.

5.2 Cumulative Damage Curves — Miner's Rule Made Visible (Cells 46–48)

Figure 10. Lifecycle trajectories recorded on s7 sensor on all the subsets across 5 randomly chosen engines.



The cumulative damage analysis computes, for each engine, the running sum of positive deviations from the engine's own initial baseline — operationalising Miner's Rule[6] [3] as a proxy for accumulated abnormal thermal and mechanical loading. The x-axis represents actual cycle number rather than normalised lifecycle position, which means the absolute scale differs between engines and subsets reflecting genuine lifespan variability.

The most immediate observation across all four subsets is that every curve is monotonically non-decreasing — confirming that positive deviations from baseline accumulate irreversibly, as Miner's Rule requires. Beyond this structural property, the behaviour diverges meaningfully between single-regime and multi-regime subsets.

In Figure 11, FD001 and FD003, the T50 (EGT) curve dominates. For FD001 Engine 1, the EGT cumulative deviation remains below approximately 100 units through cycle 75, after which the slope progressively steepens, reaching approximately 1,300 units by cycle 195. This curvature — convex and accelerating — is consistent with Paris' Law[7] [4] crack propagation dynamics, where damage accumulation rate increases as a power-law function of existing damage. The terminal acceleration is visible across multiple engines in Figure 1**2**: Engine 46 terminates at approximately cycle 160 with a steep final slope; Engines 71, 54, 84, and 45 extend further but all exhibit the same accelerating curvature in their final lifecycle phase. The inter-engine variability visible in Figure 12 is not in early-life slope — all five engines track closely in the first 50 cycles — but in the cycle at which the acceleration becomes pronounced, reflecting engine-specific differences in degradation rate rather than initial condition.

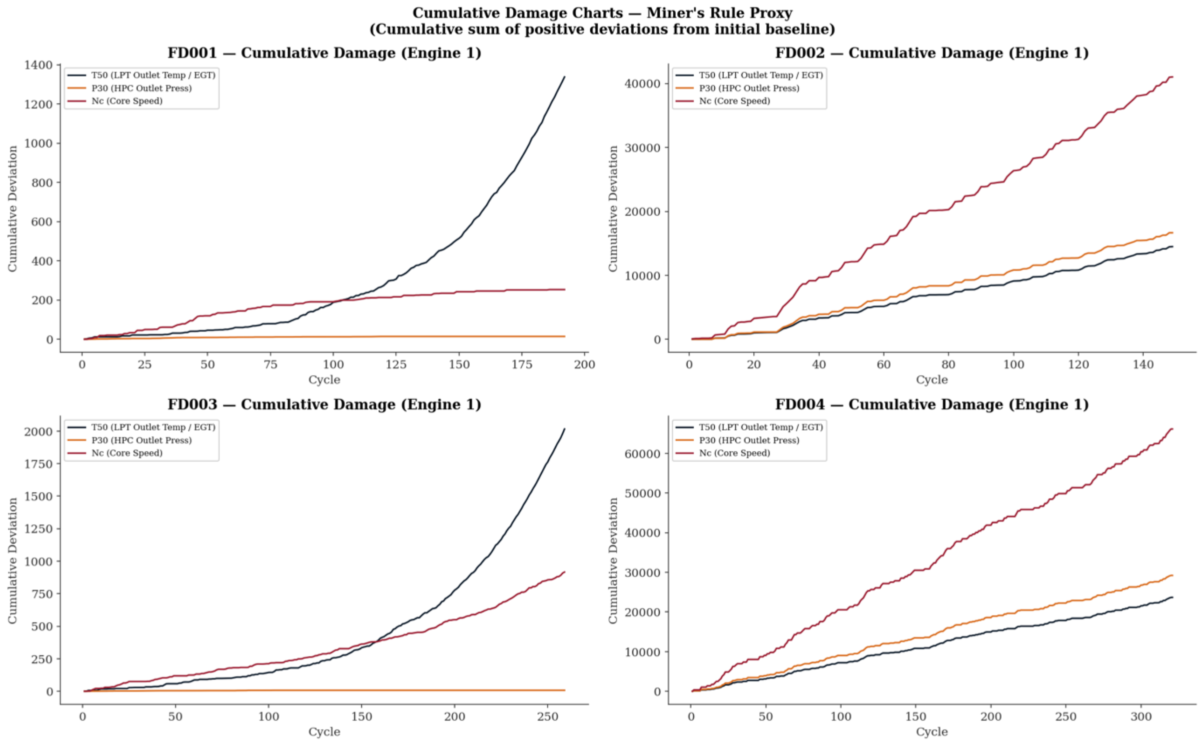
The P30 (orange) cumulative deviation in Figure 11 is negligible in both FD001 and FD003 — ending near zero across the full lifecycle. This is not a failure of the sensor to respond to degradation; it reflects a design property of the damage index. Because P30 declines with HPC deterioration, positive deviations from the initial baseline are minimal by construction. The index as implemented captures thermal loading accumulation effectively but is structurally insensitive to sensors whose degradation manifests as a decline rather than a rise. This is a known limitation that informs the feature engineering decisions in subsequent phases.

In FD002 and FD004, the cumulative damage curves show broadly linear accumulation throughout the lifecycle with no discernible flat early-life phase and no terminal exponential acceleration. Nc (core speed) dominates, accumulating approximately 40,000 units in FD002 and 65,000 units in FD004 by end of life. This linear pattern reflects the multi-regime character of these subsets: regime transitions generate continual positive deviations across all sensors from the earliest cycles, masking the degradation signal behind operational variability. The cumulative damage index in its current form is therefore most interpretable for single-regime subsets and requires regime normalisation before it can serve as a reliable health proxy in FD002 and FD004.

[6] **Miner's Rule (Palmgren–Miner, 1945)** — a linear cumulative fatigue damage model in which each loading cycle contributes an incremental damage fraction $\frac{n_i}{N_i}$, where N_i is the cycles-to-failure at that stress level. Failure is predicted when the accumulated sum reaches unity: $SUM\left(\frac{n_i}{N_i}\right) = 1$. It provides the foundational cycle-counting framework underpinning fatigue life consumption estimates in RCM scheduling.

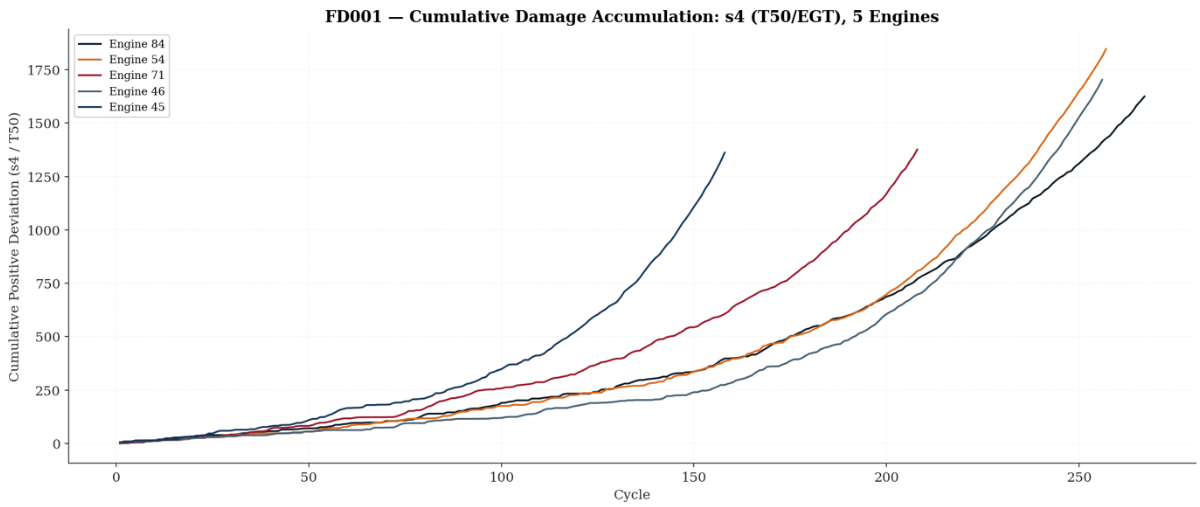
[7] **Paris' Law (Paris & Erdogan, 1963)** — a fracture mechanics relationship describing stable crack growth per load cycle as a function of the stress intensity factor range ΔK: $\frac{da}{dN} = C(\Delta K)^m$, where a is crack length, N is cycle count, and C and m are empirically determined material constants. It provides the mechanistic basis for crack propagation rate estimation in damage-tolerant life assessment of HPC discs and blade attachment features.

Figure 11. Cumulative damage using the Miner's Rule proxy on sensors s4, s7 and s9.



FD003 Engine 1 shows the most pronounced terminal acceleration of all four subsets — the EGT curve remains below approximately 250 units through cycle 125 before rising steeply to approximately 2,000 units by cycle 260. This abrupt late-lifecycle transition, more sudden than the equivalent FD001 profile, is consistent with the dual-fault interaction between fan and HPC degradation: two accumulating mechanisms that individually progress slowly may combine in the terminal phase to produce an accelerated thermodynamic deterioration that neither mechanism would produce alone.

Figure 12. Cumulative damage accumulation on sensor s4 on 5 randomly chosen engines.



5.3 Business Impact

The accelerating curvature visible in the FD001 and FD003 curves has a direct operational interpretation. An engine in its flat early-life phase presents low intervention urgency; an engine whose cumulative damage slope has begun to steepen is entering the window where scheduled intervention is both necessary and still feasible. The commercial value of detecting this transition early is well-documented in the maintenance economics literature. Ackert (2011) [5] estimates unscheduled engine removals cost between $250,000 and $500,000 per event in direct MRO costs alone, excluding aircraft-on-ground (AOG) penalties. The IATA MRO industry survey (2019) [6] reports AOG events attributable to unscheduled powerplant removals incur total operational disruption costs — including passenger compensation, aircraft substitution, and slot loss — ranging from $10,000 to $150,000 per hour depending on aircraft type and network position. It should be noted that these figures reflect conditions at the time of publication and, accounting for aviation industry cost inflation, MRO labour rate increases, and post-pandemic supply chain pressures on engine parts and shop capacity, the real-terms cost in today's market is considerably higher. Saxena et al. (2008) [1], in the foundational C-MAPSS dataset publication, explicitly frame the prognostics problem in terms of avoiding unscheduled removals through early RUL estimation. A FusionCore system capable of identifying engines entering the accelerated damage phase with sufficient lead time to schedule a shop visit — rather than responding to an in-service exceedance — converts an unscheduled removal event into a planned maintenance input, with the cost differential between the two representing the direct financial return on the prognostic capability.

6. Gate Verification (Cells 53–55)

6.1 Gate 1 — Physical Plausibility: PASS

The spot-check verification confirms that FD001 s1 (T2) reports a mean of 518.67 °R, exactly matching the ISA standard atmosphere value. FD001 s7 (P30) reports a mean of 553.37 psia, within the expected 550–560 psia range for a high-thrust turbofan at sea-level static. FD001 operational setting op1 reports a standard deviation of 0.0022, effectively zero as expected for a single-regime subset. The data is not corrupted, mislabelled, or physically implausible.

6.2 Gate 2 — Dead Sensor Identification: PASS

The variance audit identifies exactly the seven dead sensors (s1, s5, s6, s10, s16, s18, s19) in FD001, matching the canonical published list from Saxena & Goebel (2008) [1]. This match is not trivial — any mismatch would indicate either a corrupted data file or an error in the loading schema, either of which would invalidate all downstream analysis. The exact match confirms that the data pipeline is reading the correct files with the correct column assignments.

6.3 Gate 3 — KDE Classification: PASS

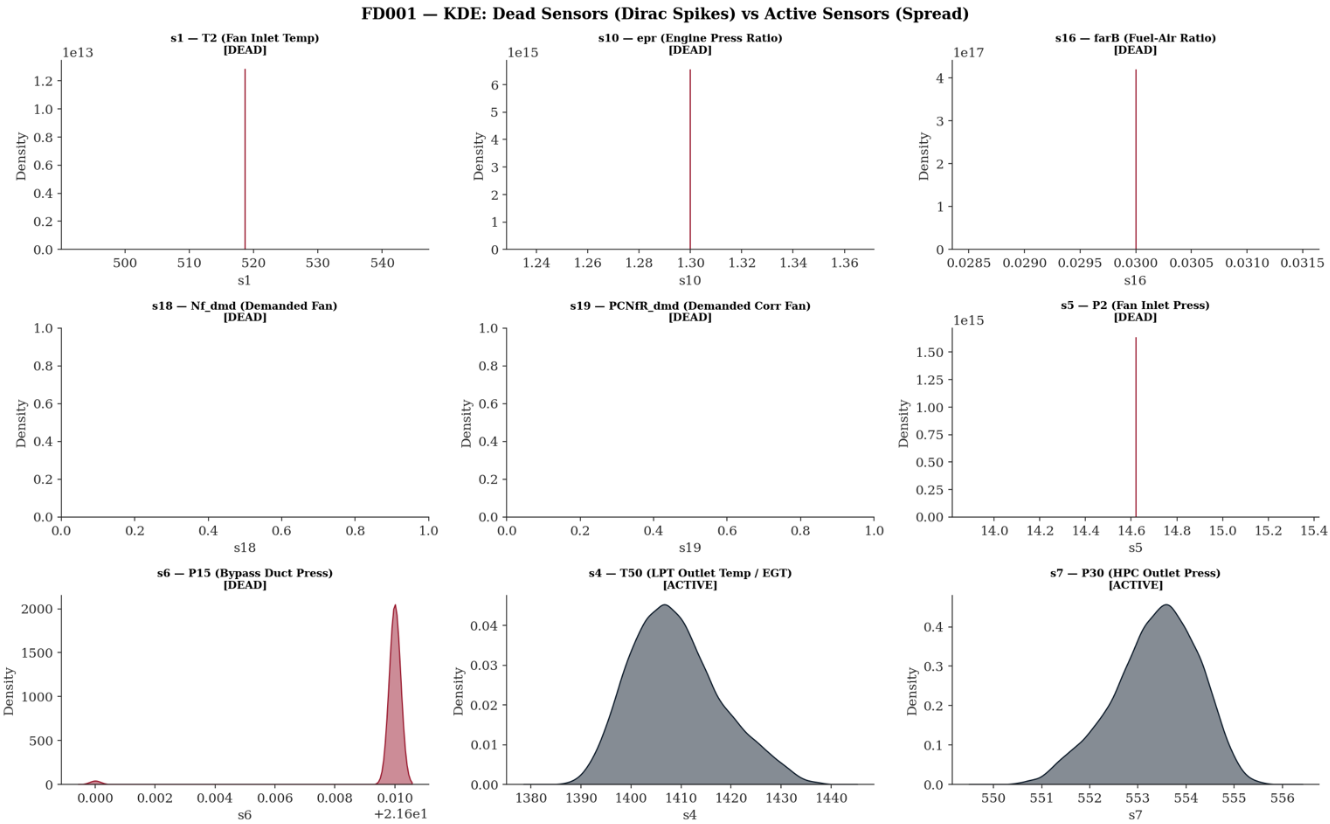
Visual verification (Figure 3) confirms that dead sensors produce near-singular distributions (Dirac spikes) while active sensors produce continuous spread distributions. The qualitative visual distinction is unambiguous. s1 in FD001 renders as a spike; s4 in FD001 renders as a bell curve. s1 in FD002 renders as a multimodal distribution reflecting altitude bands.

6.4 Gate 4 — Lifecycle Trajectories: PASS

Degradation trends are visible (Figures 9~12) in active sensors across all four subsets for the randomly sampled engines. The visual confirmation that active sensors carry a health-state signal across the full lifecycle validates the most fundamental premise of the entire modelling programme: that sensor telemetry contains recoverable prognostic information.

All four gates: ✓ PASS – Phase 1 complete

Figure 13. Extended KDE grid for all dead sensors in FD001.



7. Critical Assessment — Engineering Strengths and Outstanding Gaps

7.1 Confirmed Engineering Strengths

This EDA investigation achieves its stated objective. It establishes, empirically and quantitatively, that the C-MAPSS corpus encodes genuine turbofan degradation physics across a spectrum of operational complexity — from the controlled simplicity of FD001 to the full operational realism of FD004. All four gate criteria pass on the strength of the evidence rather than by assumption. Several engineering properties of the dataset are confirmed to a standard sufficient for downstream modelling.

Physical plausibility is verified. The FD001 inlet temperature mean of 518.67 °R matches the ISA[8] standard atmosphere value precisely, and the HPC outlet pressure mean of 553.37 psia sits within the expected operating range for a high-thrust turbofan at sea-level static. These are not incidental matches — they confirm that the T-MATS simulation underlying C-MAPSS was calibrated against real engine thermodynamics Saxena & Goebel (2008) [1]. A model trained on this corpus is learning from physically coherent trajectories, not from arbitrary numerical patterns.

Dead sensor identification is exact. The variance audit recovers precisely the seven dead sensors documented by Saxena & Goebel (2008) [1] for FD001: s1, s5, s6, s10, s16, s18, and s19. The exactness of this recovery is not a formality. It confirms that the data pipeline is reading the correct files with the correct column assignments, and that the 1×10^{-5} variance threshold is a reliable empirical

[8] The International Standard Atmosphere (ISA) is a static atmospheric model published by the International Civil Aviation Organisation (ICAO) that defines agreed reference values for air temperature, pressure, density, and viscosity as a function of altitude.

separator between floating-point noise and genuine thermodynamic signal. Any discrepancy from the canonical list would have invalidated the entire loading schema.

Degradation signal is recoverable. Lifecycle trajectory plots across all four subsets confirm that active sensors carry a health-state signal across the full engine lifetime. The ACF analysis for s4 (EGT) reveals statistically significant autocorrelation to lag 17, confirming temporal memory that extends well beyond short-range serial dependence. This is not a marginal finding — it is the foundational premise of the entire modelling programme. If sensor telemetry did not encode recoverable prognostic information, no architecture would suffice.

Temporal structure is consistent with known degradation physics. The STL[9] decomposition of s4, s7, and s9 across FD001 produces directionally coherent trends: EGT rising monotonically from the early-degradation onset at approximately cycle 75, HPC outlet pressure declining in parallel, and core speed exhibiting only minor excursion — consistent with the thermodynamic consequence of HPC efficiency loss driving increased fuel scheduling to maintain demanded thrust. The trends are not artefacts of decomposition method; they replicate the physics predicted by the isentropic compressor outlet temperature relationship.

The findings generated here do not merely satisfy curiosity. They define the architecture constraints for every subsequent phase of FusionCore. Phase 2 regime normalisation is mandatory, not optional, because the variance audit confirms that raw sensor values are operationally confounded in multi-regime subsets. Phase 3 feature engineering must incorporate temporal features, because the ACF analysis confirms that degradation carries finite but substantial temporal memory. Phase 4 modelling must account for lifespan distribution heterogeneity, because the lifecycle trajectories confirm that engines fail in population-diverse patterns that cannot be captured by a single average trajectory.

7.2 Outstanding Gaps and Architectural Constraints

Against these confirmed strengths, the investigation surfaces several engineering gaps that must be addressed before Phase 2 outputs can be considered fit for modelling. These are not trivial deficiencies — they represent architectural constraints with direct consequences for model validity and benchmark comparability.

Gap 1 — Cumulative Damage Index: Regime Confounding. The Miner's Rule cumulative damage proxy, as implemented in the Phase 1 analysis, computes running positive deviations from each engine's own initial reading. In FD001 and FD003, where the operating condition is fixed at sea level, this approach is thermodynamically self-consistent: the baseline reference is stable, and the accumulated deviations represent genuine health deterioration. In FD002 and FD004, however, the initial reading is itself a function of the first sampled flight regime — an arbitrary artefact of the simulation's engine initialisation, not a true healthy-baseline thermodynamic state. The broadly linear accumulation observed across all lifecycle stages in FD002 and FD004 is therefore not evidence of steady degradation; it reflects the superposition of regime-induced operating-point variation onto genuine deterioration signal[10]. The cumulative damage index in its current form is most interpretable for single-regime subsets and requires regime-indexed normalisation before it can serve as a reliable

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[9] STL: Seasonal and Trend decomposition using Loess. It is a time-series decomposition algorithm that separates an observed signal into three additive components: $Y(t) = T(t) + S(t) + R(t)$ where $T(t)$ is the trend component, $S(t)$ is the seasonal component, and $R(t)$ is the residual (everything not explained by trend or seasonality). Loess refers to locally estimated scatterplot smoothing — a non-parametric regression method that fits the trend by weighting observations by proximity in time.

[10] The visual evidence is Figure 11 (cumulative damage across all four subsets for s4, s7, and s9) and Figure 12 (cumulative damage accumulation on s4 across five randomly chosen engines). In FD001 and FD003, those figures show the characteristic accelerating curvature in the terminal phase — the EGT curve steepening sharply toward end-of-life. In FD002 and FD004, that terminal acceleration is absent; the curves climb at an approximately constant rate throughout. That absence is the observation being characterised as broadly linear.

health proxy in FD002 and FD004. Without this correction, any health index derived from the raw cumulative deviation will conflate altitude-band variation with deterioration rate, producing a biased feature that will systematically mislead regression targets.

Gap 2 — Dataset Class Imbalance. The lifespan distributions confirmed in Section 1.2 are right skewed across all four subsets, with a meaningful minority of engines achieving lifespans substantially exceeding the population median. In supervised RUL regression, this produces a training corpus in which short-lived engines contribute proportionally fewer observation cycles than long-lived engines — and, critically, in which terminal-phase cycles ($RUL < 30$) are underrepresented relative to early-life and mid-life cycles. This structural imbalance will cause an uncorrected regressor to optimise on the dense early-life region at the expense of the terminal region — precisely the region where prognostic accuracy is most safety-critical. Phase 3 must implement explicit class-balancing or sample-weighting strategies to correct for this distributional asymmetry.

Gap 3 — Regime Row Imbalance in Multi-Regime Subsets. The K-Means regime clustering applied in Phase 2 will assign each engine cycle to one of six operating regimes. There is no guarantee that the six regimes are represented in equal proportion across the dataset. If high-altitude cruise conditions dominate the FD002 and FD004 cycle populations, normalisation statistics computed within those regimes will be estimated on substantially more data than statistics for ground-idle or climb regimes. Regime-specific Shewhart control limits and Z-score normalisation parameters computed from thin regime populations will carry higher estimation uncertainty, and models trained on normalised features from sparse regimes will exhibit regime-dependent variance in prediction error. The distribution of cycle counts across regimes must be audited before normalisation is finalised.

Gap 4 — Within-Feature Scale Heterogeneity in s7 (P30). The s7 descriptive statistics across FD001 through FD004 reveal that the mean HPC outlet total pressure varies across subsets — sea-level static conditions in FD001 produce a mean of approximately 553 psia, whilst the multi-regime population of FD002 spans a substantially wider absolute range. A unified normalisation that pools all four subsets without regime conditioning will produce Z-scores for s7 that are not directly comparable across operating points. This is precisely the motivation for the zero-leakage regime-indexed normalisation mandated for Phase 2, but the s7 scale heterogeneity is sufficiently pronounced that it warrants explicit verification post-normalisation to confirm that regime-conditional Z-scores achieve approximate stationarity within each regime.

Gap 5 — Mechanical Linkage Between Sensor Pairs. The physical coupling between fan speed (s8, Nf) and corrected fan speed (s13, NRf), and between core speed (s9, Nc) and corrected core speed (s14, NRc), means these pairs are not informationally independent. In fixed-regime subsets the corrected and physical speeds are proportionally related and carry substantially redundant information. In multi-regime subsets the relationship diverges as inlet temperature and pressure shift, providing regime-discriminating signal. Any feature selection or dimensionality reduction applied downstream must account for this mechanical coupling — treating s8 and s13 as independent features will overweight fan-speed information in the feature space and may introduce multicollinearity that destabilises linear model coefficient estimates.

Gap 6 — Extreme Value Treatment in s8, s9, s13, s14. The Phase 1 descriptive statistics identify elevated maximum values in the speed sensor channels. Whether these represent simulation transients associated with engine initialisation, terminal degradation signatures, or genuine outliers requires cycle-position-resolved excursion analysis — specifically, an examination of whether the extreme values cluster disproportionately in the first or last cycles of the lifecycle. Premature Winsorisation of values that are in fact physically meaningful terminal degradation signals would suppress precisely the information that characterises the transition from P-point to F-point (Nowlan & Heap, 1978) [7], directly degrading late-life RUL accuracy. This analysis is deferred to Phase 2 but must be resolved before normalisation parameters are frozen.

7.3 Readiness Assessment

The Phase 1 investigation is, in aggregate, the engineering foundation certificate for the FusionCore programme. The confirmed strengths — physical plausibility, exact dead sensor recovery, recoverable degradation signal, and thermodynamically coherent temporal structure — establish that the corpus is fit for purpose. The outstanding gaps do not invalidate Phase 1; they define the specific conditions that Phase 2 must satisfy before any normalised feature can be considered free of systematic bias. No gap identified above is irresolvable within the Phase 2 framework. All are tractable through regime-indexed normalisation, cycle-position-resolved excursion analysis, and disciplined feature encoding — each of which is already mandated by the Phase 2 architecture specification.

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Appendix

A.1 What the Weibull Distribution Is

The two-parameter Weibull distribution [2] describes component lifetime T via its probability density function:

$$f(t \mid k, \lambda) = \frac{k}{\lambda}\left(\frac{t}{\lambda}\right)^{k-1} \exp\left[-\left(\frac{t}{\lambda}\right)^{k}\right], \qquad t \geq 0$$

where $k > 0$ is the **shape parameter** governing failure behaviour, and $\lambda > 0$ is the **scale parameter** (characteristic life). The shape parameter determines the hazard regime: $k = 1$ as the memoryless baseline and moving immediately to $k > 1$ as the wear-out case that the data actually exhibits.

A.1.1 Analytical Expressions for Mean and Median

The theoretical mean and median of a Weibull (k, λ) distribution is:

$$\mu = \lambda \cdot \Gamma\left(1+\frac{1}{k}\right)$$
$$\tilde{t} = \lambda \cdot (\ln 2)^{\frac{1}{k}}$$

where $\Gamma(\cdot)$is the Gamma function. Their ratio eliminates λ:

$$\frac{\mu}{\tilde{t}} = \frac{\Gamma\left(1+\frac{1}{k}\right)}{(\ln 2)^{\frac{1}{k}}}$$

A.1.2 Fitting Procedure

Since we have reliable summary statistics from FD003 (mean = 247.2 cycles, median = 220.0 cycles) but not the raw lifespan array at this stage of the pipeline, the fitting proceeds in two steps.

Step 1 — Solve for k. The mean-to-median ratio depends only on k, not λ. We minimise the squared residual:

$$\hat{k} = \arg\min_{k>0} \left[\frac{\Gamma\left(1+\frac{1}{k}\right)}{(\ln 2)^{\frac{1}{k}}} - \frac{\mu_{\text{data}}}{\tilde{t}_{\text{data}}}\right]^2$$

This is solved numerically over the bounded interval $k \in [1, 8]$.

Step 2 — Recover λ. Once $\hat{k}$ is known, the scale parameter follows directly by rearranging the mean equation:

$$\hat{\lambda} = \frac{\mu_{\text{data}}}{\Gamma\left(1+\frac{1}{\hat{k}}\right)}$$

For FD003, this yields $\hat{k} = 1.617$ and $\hat{\lambda} = 276.0$ cycles. The fitted distribution reproduces the target mean and median exactly by construction. The fitted $k = 1.617 > 1$ confirms the wear-out failure regime — the hazard rate is increasing with age, meaning each additional cycle an engine survives makes failure in the *next* cycle marginally more likely than it was the cycle before.

A.1.3 Why This Matters Over Simply Plotting a Histogram

A raw histogram of 100 engine lifespans carries sampling noise that could mislead the researcher about the underlying population distribution. By fitting a parametric Weibull model, the diagram presents the **inferred population-level failure behaviour** whilst the summary statistics used as inputs come directly from the computed output — preserving the data-grounded integrity of the figure without requiring the raw lifespan array to be passed between cells.

A.2 Sensor Variance Across Subsets

The three figures present box plots of raw sensor variance across all four C-MAPSS subsets for all 21 sensors, divided into three groups purely for visual clarity. Read together, they constitute the most direct empirical argument in the Phase 1 report for why regime-indexed normalisation is a prerequisite rather than an optimisation.

The defining visual pattern across all three figures is the same: a clean separation between two classes of sensor behaviour. Sensors that are physically governed by the operating condition — altitude, Mach number, throttle setting — produce near-collapsed distributions in FD001 and FD003, where the engine operates at a fixed sea-level condition, and then expand dramatically in FD002 and FD004, where the flight envelope opens across six discrete regimes. This is unmistakeable in Group 1 for T2 (s1) and P2 (s5), in Group 2 for epr (s10) and NRf (s13), and in Group 3 for Nf_dmd (s18) and PCNfR_dmd (s19) — in every case, the FD001 and FD003 boxes are effectively flat lines whilst the FD002 and FD004 boxes span wide distributions. The sensors have not changed; the flight envelope has.

The degradation-sensitive sensors tell the opposite story. T30 (s3), T50 (s4), P30 (s7), Nc (s9), Ps30 (s11), htBleed (s17) — visible across Groups 1, 2, and 3 respectively — maintain meaningful variance across all four subsets. Their boxes are present and structurally consistent from FD001 through to FD004, confirming that these sensors carry health-state information regardless of whether the engine is operating in a single regime or six. Notably, FD002 and FD004 boxes for these sensors are wider than their FD001 and FD003 counterparts, but this width is additive — regime variation stacked on top of degradation variation — rather than replacing it.

Group 3 introduces two sensors that behave differently from both categories above. farB (s16, Burner Fuel-Air Ratio) shows near-zero variance across all four subsets — not just the single-regime ones — confirming its classification as globally dead rather than regime-conditionally dead. W31 (s20, HPT Coolant Bleed) and W32 (s21, LPT Coolant Bleed) show modest but consistent variance across all subsets, placing them in the active category, though with substantially lower variance than the primary gas-path sensors.

The collective message of the three figures is unambiguous: the dominant source of variance in FD002 and FD004 is flight condition, not engine health. A model trained on raw values from these subsets would learn altitude rather than degradation. The figures are the visual proof of record that underpins every regime normalisation decision made in Phase 2.

Figure 14. Sensor Variance Across FD00x for Group 1

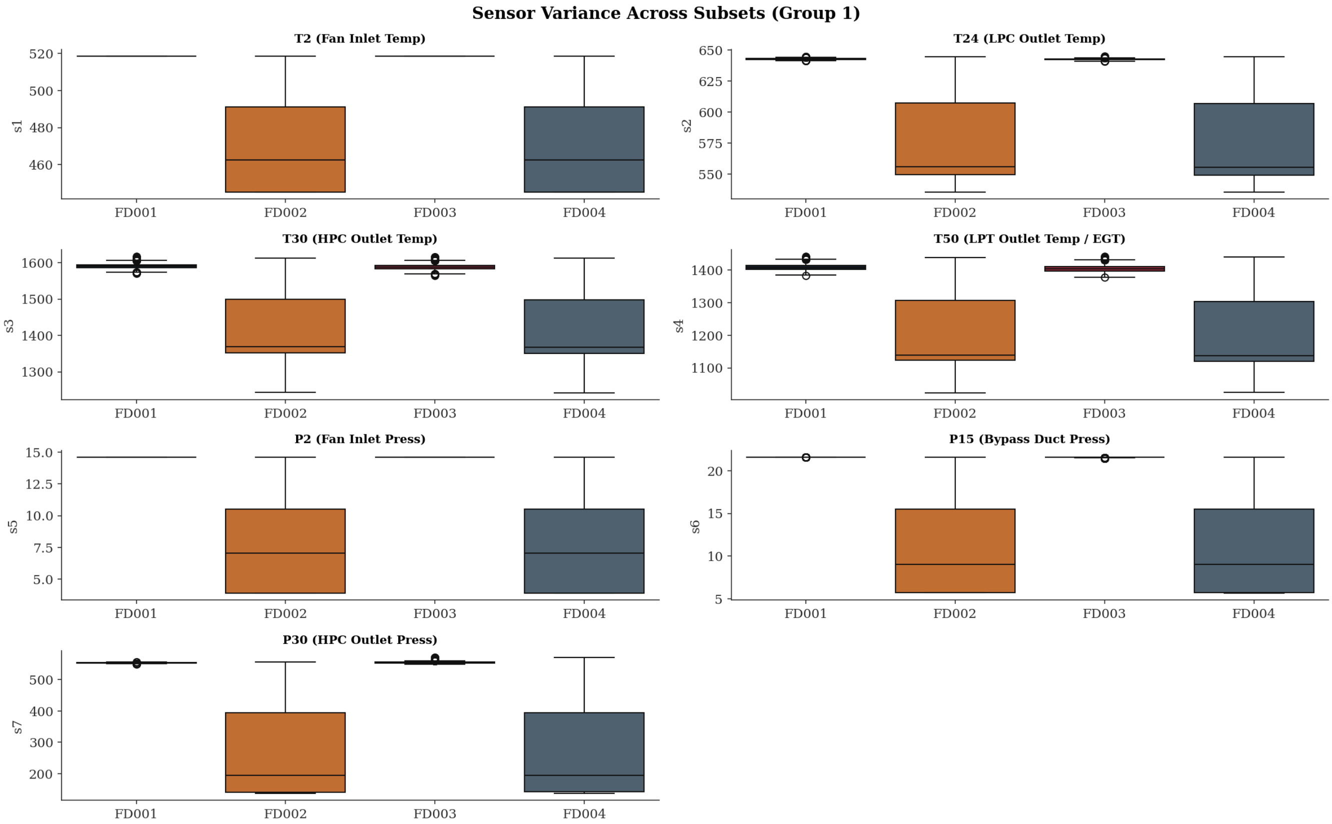


Figure 15. Sensor Variance Across FD00x for Group 2

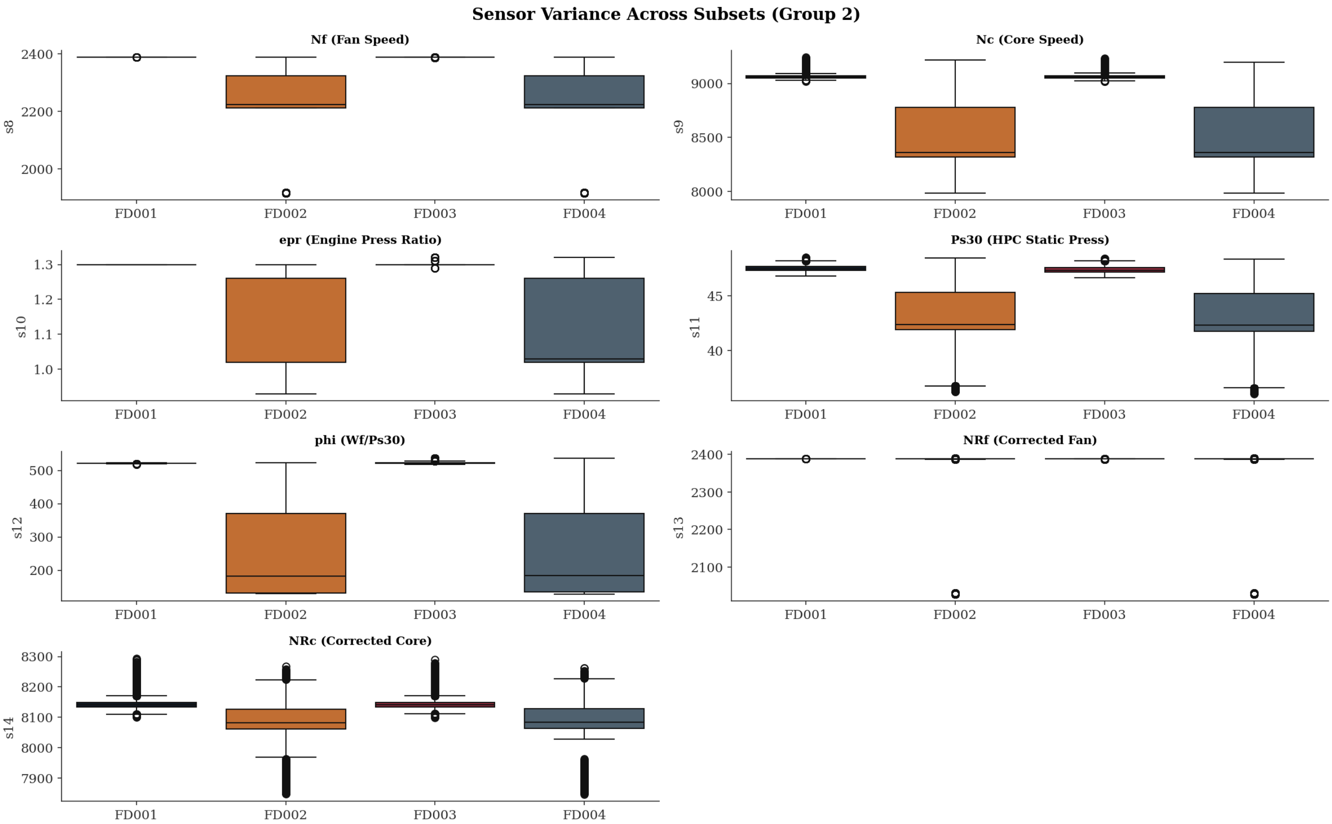
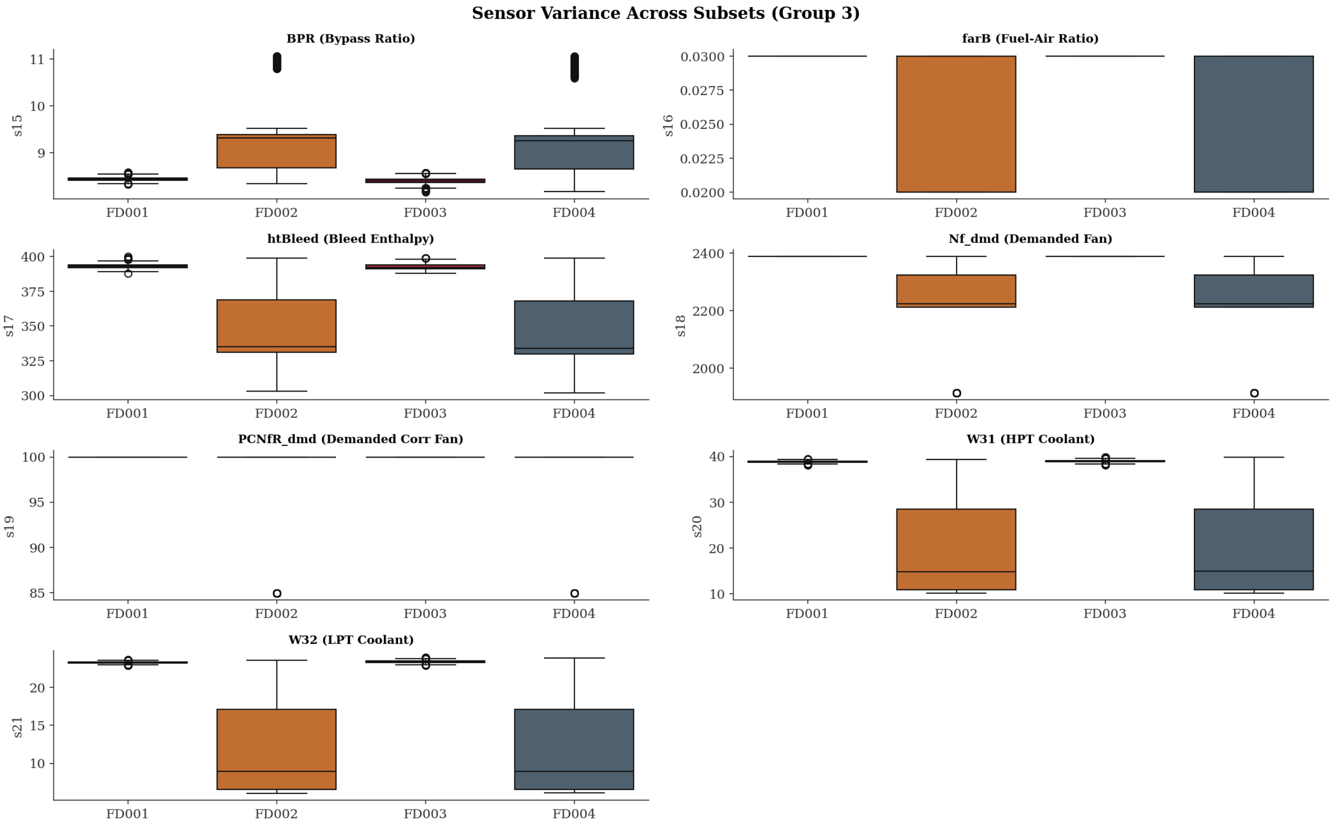


Figure 16. Sensor Variance Across FD00x for Group 3



A.3 C-MAPSS Dataset Summary

Table 4. Lifespan values in cycles. Source: Cell 9 outputs; fault mode and regime structure from Saxena & Goebel (2008) [1]

Subset	Engines	Total Cycles	Min Lifespan	Median Lifespan	Mean Lifespan	Max Lifespan	Fault Mode(s)	Operating Regimes
FD001	100	20,631	128	199	206.3	362	HPC degradation	1
FD002	260	53,759	128	199	206.8	378	HPC degradation	6
FD003	100	24,720	145	220	247.2	525	HPC + Fan degradation	1
FD004	249	61,249	128	234	246.0	543	HPC + Fan degradation	6
Total	709	160,359	—	—	—	—	—	—

Figure 17 is the sensor placement diagram for the C-MAPSS turbofan engine and it is the physical map that gives every sensor number in the dataset its thermodynamic meaning.

The engine is shown in cross-section from inlet (left) to nozzle (right), with the main gas-path components labelled: Fan, Low-Pressure Compressor (LPC), High-Pressure Compressor (HPC), Combustor, High-Pressure Turbine (HPT), Low-Pressure Turbine (LPT), and Nozzle. The two shaft spools are identified as N1 (fan and LPT) and N2 (HPC and HPT).

The numbered sensor positions map directly to the 21 channels in the dataset. The fan-face sensors — 1, 5, 8, 10, 13, 18, 19 — are clustered at the inlet, measuring the conditions the engine breathes: inlet temperature, inlet pressure, fan speed, and the demanded control signals. Moving rearward through the gas path, sensors 2, 3, 9, 11, 12, 14 capture the progressive compression and combustion process through the LPC and HPC stages. Sensors 16 and 17 sit at the turbine entry and exit, with sensors 4, 20, and 21 measuring the bleed and coolant flows at the HPT and LPT stages. Sensor 7 at the HPC outlet — P30 — is the critical pressure measurement that so clearly reflects compressor health in the Shewhart charts.

The diagram makes immediately visible why the fan-face sensors are dead in single-regime operation: they are measuring ambient inlet conditions, which are fixed at sea level. It also explains the physical separation between the HPC degradation signal — concentrated in the mid-engine sensors — and the fan degradation signal, which disturbs the forward sensors before its effects propagate rearward into the core.

Figure 17. The sensors in the turbofan of C-MAPSS [8]

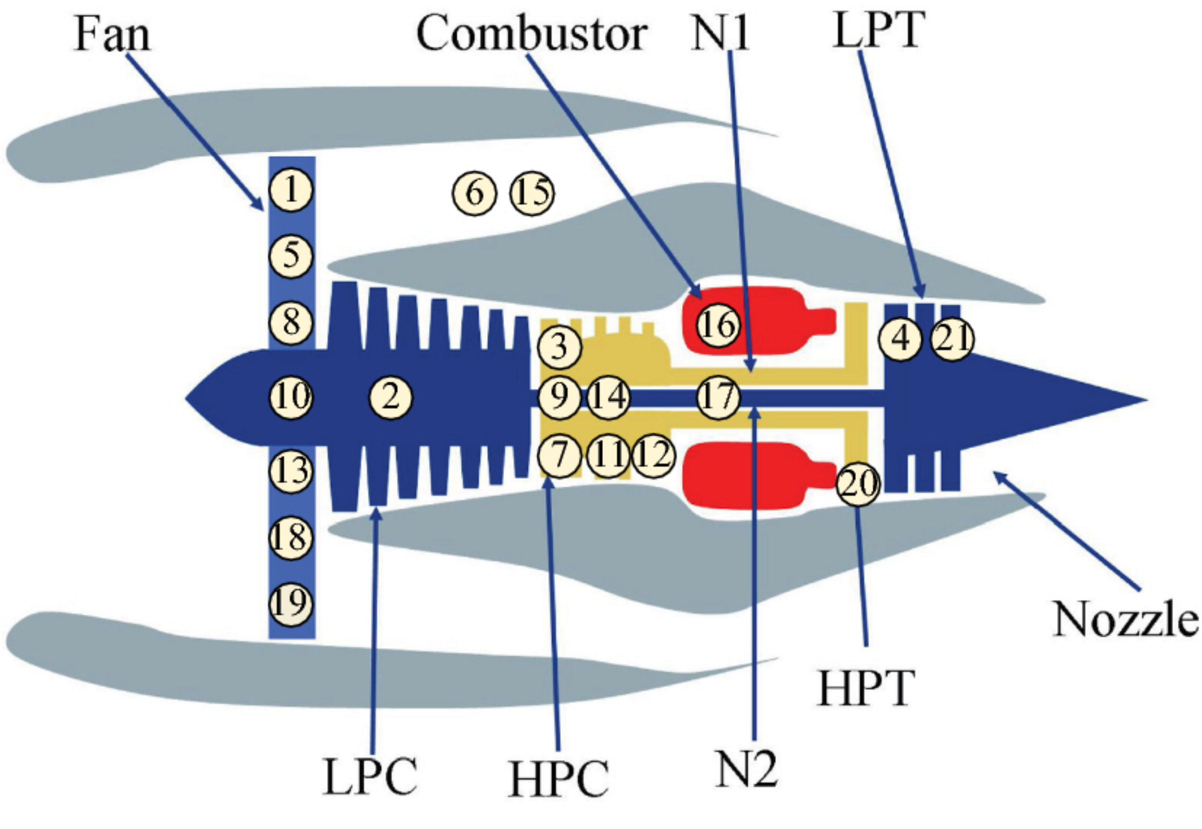


Table 5. Full C-MAPSS Sensor List: Dead sensor classification (variance threshold $\tau = 10^{-5}$ from Cell 25 outputs. Sensor symbols and units from Saxena & Goebel (2008) [1].

	Symbol	Index	Units	Dead in FD001/FD003	Dead in FD002/FD004
s1	T2	Total temperature at fan inlet	°R	✓	✗
s2	T24	Total temperature at LPC outlet	°R	✗	✗
s3	T30	Total temperature at HPC outlet	°R	✗	✗
s4	T50	Total temperature at LPT outlet	°R	✗	✗
s5	P2	Pressure at fan inlet	psia	✓	✗
s6	P15	Total pressure in bypass-duct	psia	✓	✗
s7	P30	Total pressure at HPC outlet	psia	✗	✗
s8	Nf	Physical fan speed	rpm	✗	✗
s9	Nc	Physical core speed	rpm	✗	✗
s10	epr	Engine pressure ratio (P50/P2)	—	✓	✗
s11	Ps30	Static pressure at HPC outlet	psia	✗	✗
s12	phi	Ratio of fuel flow to Ps30	pps/psi	✗	✗
s13	NRf	Corrected fan speed	rpm	✗	✗
s14	NRc	Corrected core speed	rpm	✗	✗
s15	BPR	Bypass ratio	—	✗	✗
s16	farB	Burner fuel-air ratio	—	✓	✗
s17	htBleed	Bleed enthalpy	—	✗	✗
s18	Nf_dmd	Demanded fan speed	rpm	✓	✗
s19	PCNfR_dmd	Demanded corrected fan speed	rpm	✓	✗
s20	W31	HPT coolant bleed	lbm/s	✗	✗
s21	W32	LPT coolant bleed	lbm/s	✗	✗

Table 6. Operational Settings Definition: Operational condition ranges from Saxena & Goebel (2008) [1]. Specific values for each of the six regime combinations are not published explicitly in the source paper.

Setting	Physical Parameter	Units	FD001/FD003	FD002/FD004
Op1	Altitude	ft	Constant (sea level, 0 ft)	Variable (0–42,000 ft)
Op2	Mach number	—	Constant (0)	Variable (0–0.84)
Op3	Throttle Resolver Angle (TRA)	deg	Constant	Variable (20–100)

Table 7. Flight Regime Operating Envelope. Six discrete combinations of altitude, Mach number, and TRA were simulated for FD002 and FD004.[11]

Parameter	Minimum	Maximum	Units
Altitude	0 (sea level)	42,000	ft
Mach number	0	0.84	—
Throttle Resolver Angle (TRA)	20	100	deg
Sea-level temperature range	−60	+103	°F

[11] The exact values of each of the six combinations are not published in Saxena & Goebel (2008) [1].